STATISTICAL INFERENCE

BAYESIAN INFERENCE

The Connected Teacher's Lecture

A continuous, story-driven lecture from first principles to the complete syllabus

One connected classroom narrative: concepts, derivations, examples, and all 63 examination questions

ormalsize Full spoken explanations, board narration, worked reasoning, and teaching transitions for all 63 questions.

Based on the companion study guide and its syllabus map to

Bayesian Data Analysis, Third Edition

9 November 2026

Purpose and use of this manual

This document is a companion to *Bayesian_Inference_Endsem_Guide*. Its purpose is to turn the study guide into a speech-ready teaching manual. The explanations are written as if an instructor were speaking to a class: they introduce each idea, explain why the formula is being used, narrate the algebra, interpret the result, warn about common misunderstandings, and connect the topic to the next one. The text is deliberately complete enough to be read aloud or supplied as a written lecture transcript.

The chapter order, topic headings, notation, question wording, numerical data, and intended methods follow the companion study guide. The questions are not silently replaced with different questions. For each question, the prompt is reproduced from the source manual and followed by a teacher's spoken walkthrough and the corresponding worked reasoning. Questions 1 and 2 in each type already had worked solutions; Questions 3 through 9 receive the solutions collected in the original answer key.

How to use the speech script

The instructor can read the speech passages as a continuous lecture. Equations are followed by narration explaining what each term represents and why the operation is valid. The "Board focus" panels collect the formulas that should be written or underlined during the explanation. The spoken wording is intentionally natural rather than compressed into bullet-point notes. The same blue-grey box style is used throughout for lecture emphasis, board work, and question walkthroughs.

Notation reminder. Unless the question explicitly says otherwise, the Gamma distribution uses the shape-rate convention. Thus $\text{Gamma}(a, b)$ has mean a/b and variance a/b^2. The inverse-Gamma density uses shape a and rate-like scale parameter b in the kernel $x^{-a-1}\exp(-b/x)$. Distinguish variance from precision, and distinguish rate from scale whenever you speak through a formula.

Teaching tone. The manual uses phrases such as "Let us look at this carefully" and "Notice what changes here" so that the material reads like an instructor's explanation. The aim is clarity and completeness, not a verbatim imitation of any particular person's voice.

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The story behind the entire course: one investigation, many layers of uncertainty

Opening lecture: begin with a problem, not a formula

Good morning, everyone. Today, rather than treating Bayesian inference as a collection of unrelated distributions and techniques, I want us to follow one investigation from its first question to its final scientific conclusion. Imagine that a public-health research team is studying an infectious disease. The team wants to know how common the disease is, how reliable a diagnostic test is, whether a new intervention changes the risk, how those effects differ across regions, and which measured characteristics help explain the differences.

At the beginning, the team knows very little. It has scientific background, perhaps earlier studies, and a set of assumptions, but it does not know the true disease prevalence. It collects observations, updates its beliefs, and then asks increasingly complicated questions. Each new topic in this course appears because the investigation has reached a point where the previous tool is no longer enough.

That is the story we will follow. Bayes' theorem enters because a positive test must be interpreted in light of the prevalence of disease. Conjugate distributions enter because we repeatedly observe successes, counts, waiting times, and measurements, and we want systematic posterior updates. Posterior summaries enter because a distribution must be communicated responsibly. Bayes factors enter when the team must compare competing scientific explanations. MCMC enters when the posterior becomes too complicated to calculate directly. Bayesian regression enters when several explanatory variables act together. Finally, hierarchical models enter when the team has data from many regions and wants to learn both about individual regions and about the population connecting them.

Please keep one principle in mind throughout the course: the topics are not isolated chapters. They are answers to successive questions in a single investigation. Every formula has a job. Every prior expresses a modelling decision. Every posterior is an updated account of uncertainty. Every computational method is used because the inferential problem requires it.

The map of our journey

First, we learn the language of uncertainty: parameters, data, likelihood, prior, posterior, and prediction. Next, we discover why exponential-family likelihoods and conjugate priors make updating manageable. Once we have a posterior, we learn how to summarise it without throwing away uncertainty, and how to choose priors responsibly. Then we compare hypotheses using posterior odds and Bayes factors. When exact integration becomes difficult, we learn to explore the posterior by simulation. We then move from one unknown rate to a whole regression surface, and from one population to many related groups. At the end, we return to the examination questions and see that each one is simply a particular stage of this same reasoning process.

Chapter 1

The Bayesian framework: probability, models, and updating

Story opening: the first uncertainty

Let us enter the research team's meeting room. A test kit has been developed, and the team has tested a group of people. Some tests are positive. The immediate temptation is to say that a positive result means the person has the disease. But the team knows that the test is not perfect, and the disease itself may be rare. So the first question is not merely, 'How often does the test turn positive?' The real question is, 'Given that this person's test is positive, how plausible is it that the person actually has the disease?' That question forces us to distinguish the direction of conditioning, and it brings us directly to Bayes' theorem.

Opening words for the instructor

Before we begin the first chapter, tell students that Bayesian inference is a way of reasoning under uncertainty. We will repeatedly move from a model to a posterior distribution and then to a substantive interpretation. Our first goal is to make the language precise; our second is to become fluent in the algebra; and our third is to learn how to explain what the result means in the context of the problem. Do not rush the notation. Most errors later in the course begin with a misunderstanding of what is conditional on what.

1.1 What Bayesian inference is

Teacher's spoken explanation

Let us begin with the central question of this entire course: when we do not know a quantity, how should we describe what we know about it? In a Bayesian analysis, we do not begin by pretending that one estimate tells the whole story. We describe uncertainty with a probability distribution. Before looking at the current data, we have a prior distribution. We then describe how the data would arise for each possible parameter value; that is our sampling model, and once the observed data are fixed it is also the likelihood as a function of the parameter. Finally, Bayes' rule combines those two ingredients and gives us the posterior distribution. The posterior is our updated description of uncertainty after seeing the data.

Please notice the difference between the three objects. The prior is about the unknown parameter before the current data. The likelihood is about the observed data conditional on a proposed parameter value. The posterior is about the parameter conditional on the data we actually observed. We must not swap the direction of conditioning. For example, the probability of a positive test given disease is not the probability of disease given a positive test. That reversal is

exactly why Bayes' rule is needed.
When I write the equation on the board, I want you to read it aloud as: posterior density equals prior density multiplied by likelihood, divided by the marginal probability of the observed data. The denominator makes the result integrate or sum to one. When we are deriving a posterior only up to proportionality, we can temporarily omit that denominator because it does not depend on the unknown parameter. But we must remember that the final posterior is a properly normalised probability distribution.

Board focus: equations from the companion study guide

$$p(\theta \mid y) = \frac{p(\theta)p(y \mid \theta)}{p(y)}, \qquad p(y) = \int p(\theta)p(y \mid \theta)\,\mathrm{d}\theta.$$

$$\boxed{p(\theta \mid y) \propto p(\theta)\,p(y \mid \theta).}$$

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.2 The three steps of Bayesian data analysis

Teacher's spoken explanation

A Bayesian analysis is not just one calculation. It is a three-stage process, and all three stages matter. First, we construct the full probability model. This means stating what is random, what is unknown, how observations are generated, what is assumed independent, and what prior information is being used. A model that is mathematically convenient but scientifically unreasonable is still a poor model.
Second, after the data have been observed, we condition on those data and calculate the posterior distribution. In simple conjugate problems we may obtain a closed-form expression. In a more complicated model we may need numerical integration or simulation. The computational method changes, but the target remains the posterior distribution.
Third, we evaluate the model and the consequences of the posterior. We ask whether replicated data from the model look plausible, whether the conclusions make scientific sense, and whether they change substantially under reasonable alternative prior assumptions. If not, we revise the model and repeat the process. I want to emphasise that a posterior can be calculated perfectly and still be misleading if the first-stage model is inappropriate. Computation does not repair a bad model automatically.

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.3 Parameters, data, likelihood, and predictive quantities

Teacher's spoken explanation

Let us set up the notation carefully because the same symbols will appear throughout the course. We use theta for an unknown parameter, y for the data already observed, and y-tilde for a future or otherwise unobserved quantity that could in principle be measured. A parameter is not the same thing as an observation. A future observation is also not the same thing as a parameter: it contains both the uncertainty about the parameter and the natural randomness of a new measurement.
If observations are conditionally independent given theta, their joint likelihood is the product of their individual sampling densities. The phrase 'conditionally independent' is important. It means independent after we know the shared parameter. If we average over uncertainty about that parameter, observations can become dependent marginally.
Before observing the current data, we can average the sampling distribution over the prior to obtain a prior predictive distribution. This asks: if our prior and model were true, what kind of data would we expect to see? After observing the data, we average over the posterior instead. That gives the posterior predictive distribution for a new observation. Do not simply substitute the posterior mean into the likelihood and call the result the predictive distribution. The correct predictive calculation integrates over all posterior parameter values, weighted by their posterior probabilities.

Board focus: equations from the companion study guide

$$p(y \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta).$$

$$p(\tilde{y}) = \int p(\tilde{y} \mid \theta) p(\theta) \, \mathrm{d}\theta.$$

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.4 Posterior odds and the likelihood ratio

Teacher's spoken explanation

For two competing hypotheses, probability can be expressed as odds. Odds for H-one against H-zero are the probability of H-one divided by the probability of H-zero. Bayes' rule has a particularly simple form in odds: posterior odds equal prior odds multiplied by the likelihood ratio. In a simple-hypothesis comparison, the likelihood ratio is the probability of the observed data under H-one divided by the probability of the same data under H-zero.
This is a very useful interpretation. The prior odds tell us where we started. The likelihood ratio tells us how strongly the observed data distinguish the two hypotheses. The posterior odds tell us where we finish. The data do not multiply a prior probability directly; they multiply the prior odds. After obtaining posterior odds, convert them back into probabilities by dividing the odds numerator by the sum of the numerator and denominator. For composite hypotheses, the corresponding comparison uses marginal likelihoods and is called a Bayes factor.

Board focus: equations from the companion study guide

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} \times \frac{p(y \mid H_1)}{p(y \mid H_0)}.$$

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.5 Exchangeability and conditional independence

Teacher's spoken explanation

Exchangeability means that the joint distribution of the observations does not change when we relabel their order. If I permute the observation labels, the probability model remains the same. This is a statement about symmetry, not a claim that every observed value is identical.
A common practical representation is to assume that observations are independent and identically distributed conditional on a shared unknown parameter. For example, each Bernoulli trial has the same success probability p once p is known. But when p itself is uncertain, learning from one observation changes our distribution for p, and that affects what we predict for another observation. Thus conditional independence need not imply marginal independence. This idea becomes central in hierarchical modelling, where observations within a group share group-level parameters and groups share population-level hyperparameters.

Board focus: equations from the companion study guide

$$Y_i \mid \theta \overset{\text{iid}}{\sim} p(y \mid \theta), \qquad \theta \sim p(\theta).$$

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.6 Worked examples: understanding Bayes' rule

Teacher's spoken explanation

I will use the diagnostic-test example to show why we must write the conditioning direction carefully. We are given the prevalence, the sensitivity, and the false-positive rate. The question asks for disease probability after a positive test. So the denominator must be the probability of a positive test from either route: a diseased person testing positive, or a non-diseased person testing positive. The numerator is only the first route. This gives the posterior probability. It is not equal to sensitivity because sensitivity conditions on disease, while the question conditions on the test result.
In the odds example, the arithmetic is even shorter. First convert the prior probabilities into prior odds. Then multiply by the likelihood ratio. Finally convert the posterior odds into a probability. I want students to learn the order, not just the numerical answer. If we keep the order straight,

the calculation becomes almost mechanical.

Board focus: equations from the companion study guide

$$\Pr(D, +) = 0.02(0.95) = 0.019, \qquad \Pr(D^c, +) = 0.98(0.10) = 0.098.$$
$$\Pr(+) = 0.019 + 0.098 = 0.117.$$

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

1.7 Common conceptual distinctions

Teacher's spoken explanation

Before moving on, let us separate five ideas that students often blend together. A prior is not necessarily an arbitrary guess; it may represent earlier evidence, a scientific constraint, or a population distribution. A likelihood is not a probability distribution over parameter values. A posterior is the updated distribution after conditioning on the observed data. A credible interval describes posterior probability for a parameter, whereas a predictive interval describes uncertainty about a future observation. A predictive interval is generally wider because it includes both uncertainty about the parameter and new observation-level randomness.
Finally, a proper posterior does not automatically mean a model fits well. Posterior calculation and model checking answer different questions. We can have a mathematically valid distribution under assumptions that are nevertheless inconsistent with the data. Keep this distinction in mind whenever you interpret a result.

Pause and check understanding

Pause and ask the class to explain the same idea without looking at the equation. If a student can only repeat the formula but cannot identify the prior, likelihood, posterior, and conditioning information, return to the simple example and rebuild the argument in words.

Narrative transition: from the language of Bayes to useful models

We now know how evidence changes uncertainty, but we have not yet developed a systematic collection of models for doing the updating. The research team will encounter different kinds of observations: yes-or-no test outcomes, event counts, waiting times, categorical outcomes, and continuous measurements. We need a common mathematical structure that lets us recognise these likelihoods and pair them with priors that lead to tractable posteriors. That is the role of exponential families and conjugate distributions. In the next chapter, we move from the logic of updating to the algebra of updating.

Chapter 2

Exponential families and conjugate prior distributions

Story opening: the data arrive in different forms

The public-health team now begins collecting data. In one study, each person either tests positive or negative. In another, researchers count the number of new cases in each district. In a third, they record the time until an event occurs. These data do not look identical, but their likelihoods share a mathematical structure. The exponential-family representation helps us identify the information in the sample, while conjugacy gives us a prior family that remains recognisable after observing data. We are going to see that the posterior update often amounts to adding data summaries to prior hyperparameters.

Opening words for the instructor

In this chapter we move from the general Bayes formula to families of models that can be updated exactly. The recurring pattern is to match a likelihood kernel to a prior kernel. The instructor should repeatedly ask the class: what is the unknown parameter, what is the likelihood kernel, what is the prior kernel, and which terms change after multiplication?

2.1 Why exponential families are useful

Teacher's spoken explanation

The reason we study exponential families is that they reveal a common algebraic structure behind many familiar distributions. Rather than treating Bernoulli, Poisson, exponential, and normal models as unrelated formulas, we write them in a shared form involving a natural parameter and a sufficient statistic. For independent observations, the data enter through a sum of those sufficient statistics. That is why a complicated-looking dataset can sometimes be reduced to a small number of summary quantities without losing information about the model parameter.
Conjugacy takes advantage of this structure. We choose a prior whose algebraic kernel has the same form as the likelihood kernel. After multiplication, the posterior is in the same family as the prior, with updated hyperparameters. This gives us a closed-form answer and also makes the role of prior information visible. But remember: conjugacy is a mathematical convenience. It does not prove that the model or prior is scientifically appropriate.

Board focus: equations from the companion study guide

$$p(y \mid \eta) = h(y)\exp\{\eta T(y) - A(\eta)\}.$$

$$p(y_1, \ldots, y_n \mid \eta) = \left\{\prod_{i=1}^{n} h(y_i)\right\} \exp\left\{\eta \sum_{i=1}^{n} T(y_i) - nA(\eta)\right\}.$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.2 Beta–Bernoulli and Beta–Binomial models

Teacher's spoken explanation

Here the unknown quantity is a success probability p, so it must lie between zero and one. The Bernoulli likelihood contributes one factor of p for every success and one factor of one-minus-p for every failure. If we have s successes and f failures, the likelihood kernel is p to the power s times one-minus-p to the power f.
The Beta prior has exactly the same two factors, with exponents a-minus-one and b-minus-one. When we multiply prior and likelihood, we add the exponents. That is why the posterior is Beta with parameters a plus s and b plus f. The posterior mean is the updated success-like count divided by the total updated count. The predictive probability of success on the next trial equals that posterior mean because we average the conditional success probability p over the posterior distribution. This is the simplest example of conjugate updating and should become automatic.

Board focus: equations from the companion study guide

$$p(y \mid p) \propto p^{s}(1-p)^{f}.$$

$$p \sim \mathrm{Beta}(a, b), \qquad \pi(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a, b)}, \quad 0 < p < 1.$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.3 Gamma–Poisson model, including exposure

Teacher's spoken explanation

The Poisson model is natural for counts, but be careful about what the rate means. If an observation has exposure x, its expected count is x times lambda. The exposure may represent time, population size, area, or another amount of opportunity for events. The joint likelihood contains lambda raised to the total observed count, multiplied by an exponential term involving lambda times the total exposure.
A Gamma prior in the shape-rate parameterisation has a power of lambda and an exponential

decay term. Multiplying the prior and likelihood adds the total count to the shape and adds the total exposure to the rate. The important exam trap is to update the rate with exposure, not merely with the number of observations. When each exposure equals one, the total exposure is n and we recover the familiar simpler formula. Integrating the Poisson likelihood over a Gamma prior gives a negative-binomial predictive distribution.

Board focus: equations from the companion study guide

$$L(\lambda; y) \propto \lambda^{\sum_i y_i} \exp\left(-\lambda \sum_i x_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + \sum_i y_i,\ b + \sum_i x_i\right).}$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.4 Gamma–Exponential model

Teacher's spoken explanation

The exponential distribution models positive waiting times, and here lambda is a rate. A larger rate means shorter expected waiting times because the mean is one over lambda. For n independent waiting times, the likelihood contains lambda to the power n and an exponential term involving minus lambda times the sum of the waiting times.
The Gamma prior is conjugate, so the posterior shape increases by the number of observed events and the posterior rate increases by the total waiting time. Please distinguish exact event times from right-censored observations. An exact waiting time contributes an event and its duration. A right-censored observation tells us only that the unit survived to a known time; it contributes a survival probability, which increases the total exposure but does not count an observed event. If we transform from the rate lambda to the mean waiting time, which is one over lambda, we must transform the density using the Jacobian.

Board focus: equations from the companion study guide

$$p(y \mid \lambda) \propto \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} y_i\right).$$

$$\boxed{\lambda \mid y \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^{n} y_i\right).}$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.5 Dirichlet–Multinomial model

Teacher's spoken explanation

The Beta distribution handles one probability, while the Dirichlet distribution handles a vector of category probabilities. The components must be nonnegative and add to one, so the parameter space is a simplex. In a multinomial sample, the likelihood kernel is the product of each category probability raised to its observed count.
The Dirichlet prior has the same product structure, with each exponent equal to its concentration parameter minus one. Multiplying adds the observed category count to the corresponding prior concentration parameter. The posterior is therefore Dirichlet with updated component-wise parameters. The posterior mean for category k is its updated concentration divided by the sum of all updated concentrations. The predictive probability that the next observation belongs to category k is exactly this posterior mean. If there are only two categories, the Dirichlet model reduces to the Beta model.

Board focus: equations from the companion study guide

$$p(y \mid \boldsymbol{p}) \propto \prod_{k=1}^{K} p_k^{n_k}.$$

$$\boxed{\boldsymbol{p} \mid y \sim \text{Dirichlet}(\alpha_1 + n_1, \ldots, \alpha_K + n_K).}$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.6 Normal likelihood with a conjugate Normal prior: known variance

Teacher's spoken explanation

In this model, the sampling variance is known, and the unknown parameter is the population mean. The sample mean contains the relevant information about the mean. We assign a normal prior to the unknown mean. The likelihood precision is n divided by sigma-squared, while the prior precision is one divided by the prior variance.
When we multiply the two normal kernels and complete the square, the posterior is normal. Its precision is the sum of prior precision and data precision. Its mean is a weighted average of the prior mean and the sample mean, with weights proportional to precision. This is why we say the posterior is a compromise between prior information and data. Strong prior precision gives greater influence to the prior mean; a large sample or small known sampling variance gives greater influence to the sample mean.

Board focus: equations from the companion study guide

$$\mu \mid y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\boxed{\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_n = \tau_n^2\left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}\right).}$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.7 Normal–Gamma and Normal–Inverse-Gamma conjugacy

Teacher's spoken explanation

Now both the mean and variance are unknown, so the prior has to describe their relationship as well. In the Normal–Gamma version, we work with the precision tau, which is one over sigma-squared. Conditional on tau, the mean is normal, and the precision itself has a Gamma prior. In the Normal–Inverse-Gamma version, we use sigma-squared directly and give it an inverse-Gamma prior.
The posterior remains in the same family. The effective prior sample size parameter kappa increases by n. The posterior mean is the weighted average of the prior mean and sample mean. The Gamma shape or inverse-Gamma shape increases by n over two. The updated scale parameter includes the within-sample sum of squares and a correction for the difference between the sample mean and prior mean. That last term matters: it records the disagreement between the data location and the prior location. In an exam, state whether your Gamma distribution uses rate or scale and whether the prior is on variance or precision.

Board focus: equations from the companion study guide

$$\tau \sim \text{Gamma}(a_0, b_0), \qquad \mu \mid \tau \sim \mathcal{N}\left(\mu_0, \frac{1}{\kappa_0 \tau}\right), \qquad Y_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1}).$$

$$\tau \mid y \sim \text{Gamma}(a_n, b_n), \qquad \mu \mid \tau, y \sim \mathcal{N}\left(\mu_n, \frac{1}{\kappa_n \tau}\right),$$

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

2.8 Exponential-family summary table

Teacher's spoken explanation

This table is intended to be used as a quick recognition exercise. When you see successes and failures, think Beta–Binomial. When you see counts and exposure, think Gamma–Poisson. When you see waiting times and a rate, think Gamma–Exponential. When you see category counts, think Dirichlet–Multinomial. When you see normal observations with a known variance and unknown mean, think Normal–Normal. When both the mean and variance are unknown, think Normal–Gamma or Normal–Inverse-Gamma.

Do not memorise only the final update. Train yourself to recognise the likelihood kernel. Then write the prior kernel and multiply. If you can do those steps, you can reconstruct the posterior even if you forget a formula.

Pause and check understanding

Ask students to state the likelihood kernel before naming the conjugate prior. Then ask them to show which exponents are added. This small verbal check catches most shape-rate and pseudo-count mistakes before they spread into the final answer.

Narrative transition: a posterior is not yet a conclusion

The team can now calculate posterior distributions for several basic models. But suppose the director asks, 'What is our best estimate of prevalence?' Or, 'Can we give a range that expresses uncertainty?' A posterior distribution contains all the information, but people often need a concise summary. The next chapter teaches us how to extract a point estimate, a credible interval, and a defensible prior specification without confusing a summary with the full posterior.

Chapter 3

Bayesian estimation, MAP, credible intervals, and prior choice

Story opening: communicating what has been learned

Imagine that the team has observed the test outcomes and has obtained a posterior distribution for disease prevalence. The data analyst is asked to present one number at a meeting. But a single number cannot tell the whole story. A posterior mean, median, or mode answers a different decision question; a credible interval communicates a range of plausible parameter values; and the prior determines how earlier information was combined with the sample. We therefore need to learn not only how to calculate a summary, but why that summary is appropriate and how sensitive it is to prior assumptions.

Opening words for the instructor

We now turn from deriving a posterior to summarising it. Emphasise that the posterior distribution is the result; the mean, median, mode, and intervals are different ways of reporting aspects of that result. The chosen summary must match the decision problem and the question being asked.

3.1 Posterior distributions contain more than a point estimate

Teacher's spoken explanation

After the data have been observed, the posterior distribution is our complete representation of uncertainty under the model. A point estimate is a summary, and the choice of summary depends on the question or loss function. Under squared-error loss, the posterior mean is optimal. Under absolute-error loss, a posterior median is optimal. The posterior mode, or MAP estimate, is the value at which the posterior density is highest.

The MAP estimate is obtained by maximising the log likelihood plus the log prior, because the normalising constant does not depend on the parameter. If the prior is flat on the chosen parameter scale, the posterior mode and maximum-likelihood estimate coincide, assuming a maximum exists. But a density mode can change after a nonlinear transformation because the transformed density includes a Jacobian. Therefore, do not treat mean, median, and mode as interchangeable, and do not assume MAP is invariant to parameterisation.

Board focus: equations from the companion study guide

$$\hat{\theta}_{\text{Bayes}} = \mathbb{E}(\theta \mid y).$$

$$\boxed{\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(\theta \mid y) = \arg\max_{\theta} \{\log p(y \mid \theta) + \log p(\theta)\}.}$$

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

3.2 Posterior variance and the value of data

Teacher's spoken explanation

The law of total variance tells us how prior uncertainty is divided. One part is the expected posterior variance, which represents uncertainty remaining after observing data. The other part is the variance of the posterior mean across possible datasets, which represents how much the data can move our estimate. Their sum is the prior variance.
This identity implies that posterior variance is no greater than prior variance on average over the prior predictive distribution. It does not say that every particular dataset must reduce uncertainty. If the observed data conflict with a very concentrated prior, the posterior can actually be more spread out than the prior. So when you explain this theorem, use the phrase 'on average over possible data' rather than saying 'the posterior variance is always smaller'.

Board focus: equations from the companion study guide

$$\mathbb{E}(\theta) = \mathbb{E}\{\mathbb{E}(\theta \mid Y)\},$$

$$\boxed{\mathrm{Var}(\theta) = \mathbb{E}\{\mathrm{Var}(\theta \mid Y)\} + \mathrm{Var}\{\mathbb{E}(\theta \mid Y)\}.}$$

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

3.3 Credible intervals

Teacher's spoken explanation

A credible interval is a set of parameter values that contains a specified amount of posterior probability. For a central 95 percent interval, we take the 2.5th and 97.5th posterior percentiles. The interpretation is direct: conditional on the observed data and the model, the posterior probability that the parameter lies in that interval is 95 percent.
A highest posterior density region also contains 95 percent posterior mass, but it gathers the points with greatest density first. For a symmetric unimodal posterior, the central interval and HPD interval agree. For a skewed or multimodal posterior, they may differ, and the HPD set may even consist of separate pieces. The central interval is often easier to calculate from posterior simulations and has a clear quantile definition. The HPD region depends on the parameter scale, so it generally does not stay the same under nonlinear transformations.

Board focus: equations from the companion study guide

$$\Pr\{\theta \in C(y) \mid y\} = 1 - \alpha.$$

$$\left[q_{\alpha/2}, q_{1-\alpha/2}\right],$$

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

3.4 Informative, weakly informative, and noninformative priors

Teacher's spoken explanation

A prior can be informative when it contains substantial prior knowledge about the quantity, weakly informative when it rules out implausible values but still allows a broad range, or noninformative in an attempt to let the likelihood dominate. These labels are not absolute. A prior that is weak on one scale may be highly informative on another scale.
A flat prior is not automatically neutral. Flatness depends on parameterisation, and an improper prior has an arbitrary multiplicative constant because it cannot be normalised to integrate to one. Such priors can sometimes produce proper posteriors for parameter estimation, but they require care. The practical lesson is to choose priors with the actual scale and scientific meaning in mind, check the prior predictive implications, and examine sensitivity of conclusions to reasonable alternatives.

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

3.5 Jeffreys' prior

Teacher's spoken explanation

Jeffreys' prior is constructed from the Fisher information. For a scalar parameter, we calculate the expected negative second derivative of the log likelihood, and then take its square root. The result is used as a prior density up to proportionality. Its central motivation is invariance under smooth one-to-one reparameterisation.
For the binomial probability p, the Fisher information is n divided by p times one-minus-p. Taking the square root gives a density proportional to p to the minus one-half times one-minus-p to the minus one-half. That is a Beta one-half, one-half prior. It differs from the uniform Beta one, one prior. Jeffreys' prior is not a magic answer for all models: in multiparameter problems it can be controversial, and the resulting posterior still has to be checked for propriety.

Board focus: equations from the companion study guide

$$I(\theta) = \mathbb{E}\left[\left\{\frac{\partial}{\partial\theta}\log p(Y \mid \theta)\right\}^2 \middle| \theta\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(Y \mid \theta) \middle| \theta\right],$$

$$\boxed{\pi_J(\theta) \propto \sqrt{I(\theta)}.}$$

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

3.6 Prior sensitivity and transformations

Teacher's spoken explanation

Prior sensitivity asks whether substantive conclusions change when we replace the chosen prior by other reasonable prior distributions. This is not a search for a prior that gives the answer we prefer. It is a way of measuring how strongly the conclusion depends on assumptions that are not determined by the current data alone.
When transforming a parameter, transform the density, not just the variable name. If eta is a function of theta, the density of eta contains the absolute derivative of theta with respect to eta. This Jacobian can make a flat prior on one scale non-flat on another. The same issue affects modes and HPD intervals. Whenever a question asks for an induced prior, write the transformation, calculate the inverse transformation, differentiate it, and multiply by the absolute Jacobian.

Board focus: equations from the companion study guide

$$p_\phi(\phi) = p_\theta(h^{-1}(\phi))\left|\frac{\mathrm{d}h^{-1}(\phi)}{\mathrm{d}\phi}\right|.$$

Pause and check understanding

Ask: what does this summary optimise or guarantee? The answer should distinguish posterior mean, median, mode, central quantiles, and highest-density regions. Encourage students to state the interpretation before giving the number.

Narrative transition: estimation becomes a scientific comparison

The team has estimated the prevalence and described its uncertainty. The next meeting is more demanding. A policymaker asks whether the intervention truly reduces disease risk, or whether the apparent difference could be explained by sampling variation. This is no longer just a question about estimating one parameter. It is a comparison between scientific hypotheses. Bayesian hypothesis testing uses posterior odds and Bayes factors to quantify how the observed data change the relative support for competing models.

Chapter 4

Bayesian hypothesis testing and Bayes factors

Story opening: two explanations compete

The research team now has two possible explanations. Under the first, the intervention has no effect. Under the second, it changes the disease risk. We should not decide between them by looking only at which model can fit the observed data most closely. A flexible model can often fit well simply because it has more room to move. Bayesian model comparison asks how well each model predicted the data before those data were observed, averaging over the parameter values allowed by the model. This is why the marginal likelihood and the Bayes factor become central.

Opening words for the instructor

This chapter treats hypotheses as competing probability models. The central calculation is the Bayes factor, but the interpretation requires prior model probabilities and proper parameter priors. Keep the model labels consistent on the board so that no one reverses the numerator and denominator.

4.1 Simple and composite hypotheses

Teacher's spoken explanation

In Bayesian hypothesis testing, a hypothesis is treated as a model with a prior probability. A simple hypothesis specifies the parameter completely. A composite hypothesis leaves some parameter values unknown and assigns a prior distribution over them. For a simple hypothesis, the probability of the data under that hypothesis is obtained directly from the sampling model. For a composite hypothesis, we average the likelihood across the parameter prior to obtain a marginal likelihood.

This distinction explains why the Bayes factor is not always the ratio of two likelihood values evaluated at best-fitting parameters. It compares the probability of the observed data under the complete models, including their prior distributions. We must specify prior model probabilities as well as parameter priors before the posterior model probabilities can be calculated.

Board focus: equations from the companion study guide

$$\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)} = \frac{\Pr(H_1)}{\Pr(H_0)} BF_{10},$$

$$\boxed{BF_{10} = \frac{p(y \mid H_1)}{p(y \mid H_0)}.}$$

Pause and check understanding

Ask students to identify the numerator model and denominator model. Have them state whether the parameter is fixed or integrated over a prior. Then ask what additional information is needed to convert a Bayes factor into a posterior probability.

4.2 Interpreting a Bayes factor

Teacher's spoken explanation

The Bayes factor for model one against model zero is the marginal likelihood under model one divided by the marginal likelihood under model zero. It tells us how the observed data change the odds between the models. Posterior odds equal prior odds multiplied by the Bayes factor.
A Bayes factor greater than one favours the numerator model relative to the denominator model, while a value less than one favours the denominator model. But the strength of evidence is conditional on the models and their priors. A Bayes factor does not by itself give the posterior probability of a hypothesis. To obtain that probability, we also need the prior model odds.

Pause and check understanding

Ask students to identify the numerator model and denominator model. Have them state whether the parameter is fixed or integrated over a prior. Then ask what additional information is needed to convert a Bayes factor into a posterior probability.

4.3 Point null hypotheses and continuous alternatives

Teacher's spoken explanation

A point null hypothesis says that a parameter has exactly one specified value, such as theta equals zero. A continuous alternative assigns probability over a range of values. Under the null, we evaluate the likelihood at the fixed value. Under the alternative, we integrate the likelihood over the prior distribution for theta.
This integral is the key difference. The alternative is rewarded when a substantial portion of its prior distribution predicts data like those observed. A very flexible alternative may fit the data extremely well at one narrow point but assign most prior mass to values that predict the data poorly. Its marginal likelihood can therefore be smaller than expected. This is one reason Bayes factors naturally penalise diffuse alternatives.

Pause and check understanding

Ask students to identify the numerator model and denominator model. Have them state whether the parameter is fixed or integrated over a prior. Then ask what additional information is needed to convert a Bayes factor into a posterior probability.

4.4 Sensitivity, improper priors, and limitations

Teacher's spoken explanation

Bayes factors require marginal likelihoods, and marginal likelihoods require proper normalised priors or a carefully constructed alternative. With an improper prior, the arbitrary normalising constant does not cancel in general when comparing models. Consequently, the ordinary Bayes factor may be undefined.
Bayes factors can also be highly sensitive to prior scales under composite alternatives. Before interpreting a Bayes factor, report the model priors and the parameter priors, and consider how the answer changes under reasonable alternatives. A large Bayes factor is not a universal statement independent of modelling choices. It is evidence in favour of one fully specified model relative to another.

Board focus: equations from the companion study guide

$$BF_{10}(y_1, y_2) = BF_{10}(y_1)BF_{10}(y_2),$$

Pause and check understanding

Ask students to identify the numerator model and denominator model. Have them state whether the parameter is fixed or integrated over a prior. Then ask what additional information is needed to convert a Bayes factor into a posterior probability.

Narrative transition: the posterior becomes too complicated to write down

For the simple models we have studied, integration was manageable and the posterior could be written in a familiar closed form. But the intervention study may contain many parameters, dependencies, and latent quantities. The posterior still exists as a mathematical object, yet we may no longer be able to evaluate its normalising constant or calculate its moments by hand. We therefore need a way to explore the posterior numerically. This brings us to Markov chain Monte Carlo: instead of solving every integral exactly, we construct a simulation whose long-run behaviour represents the posterior distribution.

Chapter 5

Posterior simulation: Gibbs sampling, Metropolis–Hastings, and diagnostics

Story opening: when exact calculation is no longer practical

The team has expanded the model. There are now several unknown parameters, and each parameter depends on the others through the likelihood and prior. We can describe the joint posterior, but direct integration is too difficult. Think of MCMC as a carefully designed journey through the parameter space. The algorithm visits values more often in regions with greater posterior probability. Gibbs sampling does this by repeatedly drawing from full conditional distributions; Metropolis–Hastings proposes moves and accepts or rejects them with a probability that preserves the target posterior. But a long simulation is not automatically a trustworthy one. We must diagnose mixing, convergence, and effective information.

Opening words for the instructor

We now enter computational Bayesian inference. In simple conjugate models we can write the posterior directly; in complex models we often need simulation. The goal of MCMC is to produce draws from the posterior, but the instructor must explain that the draws are dependent and that convergence and mixing have to be assessed.

5.1 Why MCMC is required

Teacher's spoken explanation

In many Bayesian models, the posterior is known only up to a proportionality constant, or it is too complicated to integrate directly. Markov chain Monte Carlo allows us to generate dependent draws whose long-run distribution is the posterior target. We can then approximate expectations, quantiles, probabilities, and predictions using those draws.

We must remember what MCMC does and does not do. It does not make the model correct. It does not automatically guarantee that the algorithm has explored the posterior. It gives us a computational route to posterior summaries, provided the chain has reached the relevant stationary distribution and mixes adequately. That is why convergence diagnostics are part of the statistical analysis, not optional decoration.

Board focus: equations from the companion study guide

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \ldots, \theta^{(1)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

5.2 Gibbs sampling

Teacher's spoken explanation

Gibbs sampling is especially natural when we can sample from full conditional distributions. Suppose the unknown parameter consists of several blocks. We hold all blocks except the first fixed and draw the first block from its conditional posterior. Then we update the second block conditional on the newly updated first block and the remaining current values, and continue through all blocks.
After one complete sweep, we have a new joint state. Repeating this process creates a Markov chain with the desired joint posterior as its stationary distribution under the usual conditions. The essential exam task is to derive the full conditional distributions correctly and state the update order. Remember that the chain draws are dependent: the next draw is based on the current state. This dependence is expected and must be considered when assessing effective sample size.

Board focus: equations from the companion study guide

$$\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$
$$\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \ldots, \theta_d^{(t-1)}, y),$$

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

5.3 Metropolis–Hastings

Teacher's spoken explanation

Metropolis–Hastings is useful when direct sampling from the target posterior is difficult but we can evaluate its density up to a constant. From the current state, propose a candidate using a proposal density. Then calculate an acceptance probability that compares the target density at the candidate and current state, while correcting for asymmetry in the proposal mechanism.
The correction is the reverse proposal probability divided by the forward proposal probability. If the proposal is symmetric, those terms cancel. We accept the candidate with the calculated probability; otherwise we remain at the current state. A common mistake is to omit the proposal ratio for an asymmetric proposal. Another mistake is to think that a candidate with lower posterior density must always be rejected. It may still be accepted, which allows the chain to explore the full target distribution.

Board focus: equations from the companion study guide

$$\boxed{\alpha(\theta, \theta^*) = \min\left\{1, \frac{q(\theta^*)g(\theta \mid \theta^*)}{q(\theta)g(\theta^* \mid \theta)}\right\}.}$$

$$\alpha = \min\left\{1, \frac{q(\theta^*)}{q(\theta)}\right\}.$$

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

5.4 Convergence: what must be checked

Teacher's spoken explanation

A chain that looks busy is not necessarily converged. We want evidence that it has settled into the target distribution and is moving around that distribution effectively. Trace plots show the sequence over iterations. If there is a trend, long drift, or chains in different regions, that is a warning. Overlaid marginal density plots show whether separate chains are exploring similar distributions. Autocorrelation plots show how strongly nearby draws depend on each other.
Use several chains with dispersed starting points. If every chain begins at the same convenient point, we may fail to notice that the sampler has not explored another important region. Diagnostics should be applied to all important parameters and derived quantities, not only to one easy parameter. Convergence assessment is a collection of graphical and numerical checks rather than a single pass/fail test.

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

5.5 Scale reduction and effective sample size

Teacher's spoken explanation

The scale-reduction diagnostic compares variation within chains with variation between chains. If chains have explored the same target distribution, those sources of variation should become similar. A value close to one is reassuring, but it is not a proof of convergence. We still need to inspect the plots and the scientific behaviour of the chain.
Effective sample size adjusts the raw number of draws for autocorrelation. If successive draws are highly correlated, ten thousand saved draws may contain much less than ten thousand independent-draws worth of information. The integrated autocorrelation time is one plus twice the sum of positive-lag autocorrelations, and effective sample size is approximately the raw sample size divided by that factor. The Monte Carlo standard error of a posterior mean is approximately the posterior standard deviation divided by the square root of effective sample size. Do not confuse Monte Carlo error with posterior uncertainty.

Board focus: equations from the companion study guide

$$B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_j - \bar{\theta}_\cdot)^2,$$

$$W = \frac{1}{m}\sum_{j=1}^{m} s_j^2.$$

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

5.6 Warm-up, thinning, and practical reporting

Teacher's spoken explanation

Warm-up is the initial phase used to let the chain move away from its starting value and adapt if the algorithm requires adaptation. We do not include warm-up draws in posterior summaries. But deleting a fixed number of draws does not prove that the remaining chain has converged; diagnostics must support that decision.
Thinning means retaining every kth draw. It can reduce storage, but it does not create information and usually is not necessary merely to reduce autocorrelation. In a report, describe the sampler, number of chains, starting values, warm-up, retained iterations, convergence diagnostics, effective sample sizes, and Monte Carlo standard errors. The reader must be able to judge whether the reported summaries are based on reliable exploration of the posterior.

Pause and check understanding

Ask what the chain's state is, what distribution it is targeting, and what evidence would show that the run is reliable. A good answer should mention multiple chains, graphical diagnostics, autocorrelation, and effective sample size where relevant.

Narrative transition: from one parameter to a system of predictors

So far, our scientific questions have often involved one rate or a small collection of parameters. In practice, disease risk may depend on age, exposure, treatment, socioeconomic conditions, and region at the same time. The research team now needs a model that describes how an outcome changes with several predictors. Linear regression provides the likelihood structure, and Bayesian priors provide a way to stabilise estimates, quantify uncertainty, and express beliefs about plausible coefficient sizes. We now move from basic Bayesian models to Bayesian regression.

Chapter 6

Bayesian linear models, ridge, LASSO, and spike-and-slab

Story opening: explaining variation rather than only counting it

The research team has moved beyond counting cases. It wants to explain why risk differs between individuals. Let the response be represented by a vector and let the columns of a design matrix record the predictors. The regression coefficients describe the relationship between those predictors and the outcome. Once we place a prior on the coefficients and the error variance, we obtain a Bayesian linear model. The prior is not an optional decoration: it can stabilise estimates, especially when predictors are numerous or correlated. Different prior choices lead to ridge-type shrinkage, LASSO-type sparsity, or explicit uncertainty about which variables belong in the model.

Opening words for the instructor

This chapter connects Bayesian priors to regression regularisation and variable selection. Move carefully from the regression likelihood to the prior, then to the posterior objective. The three methods ridge, LASSO, and spike-and-slab all constrain or regularise coefficients, but they do so in mathematically different ways.

6.1 The Gaussian linear regression model

Teacher's spoken explanation

In Bayesian linear regression, the response vector is modelled as a normal distribution centred at X beta. The design matrix contains the predictors, and beta contains the regression coefficients. The likelihood describes the data conditional on beta and the residual variance. Bayesian analysis adds a prior distribution to the coefficients and, when the variance is unknown, a prior to the variance as well.
The posterior is the combination of the regression likelihood and these priors. A flat prior on beta gives the familiar least-squares centre under standard rank conditions, while an informative prior regularises coefficients and can stabilise problems with many predictors. The important habit is to distinguish the design matrix, the coefficient vector, and residual variance, and keep dimensions consistent when completing the square.

Board focus: equations from the companion study guide

$$y \mid \beta, \sigma^2 \sim \mathcal{N}_n(X\beta, \sigma^2 I_n).$$

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{ -\frac{(y - X\beta)^\top (y - X\beta)}{2\sigma^2} \right\}.$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.2 Normal–Inverse-Gamma prior and posterior

Teacher's spoken explanation

The Normal–Inverse-Gamma prior couples the coefficient vector and variance. Conditional on sigma-squared, beta is multivariate normal. The variance itself is inverse-Gamma. This prior is conjugate for the Gaussian linear regression likelihood, so the posterior has the same family.
To derive the update, combine the residual quadratic form from the likelihood with the prior quadratic form for beta. Collect the coefficient terms to identify the posterior precision matrix and posterior mean. Then complete the square. The remaining terms update the inverse-Gamma shape and rate. I want students to show the matrix algebra, because simply quoting the posterior parameter names does not demonstrate why the update is correct.

Board focus: equations from the companion study guide

$$\beta \mid \sigma^2 \sim \mathcal{N}_p(b_0, \sigma^2 V_0), \qquad \sigma^2 \sim \text{InvGamma}(a_0, d_0),$$
$$V_n = (V_0^{-1} + X^\top X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^\top y),$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.3 Posterior prediction in linear regression

Teacher's spoken explanation

Prediction requires one more step beyond estimating the regression coefficients. For a new predictor vector x-star, the conditional mean response is x-star transpose beta. But beta is uncertain, and sigma-squared may also be uncertain. Therefore the predictive distribution averages the sampling distribution of the new response over the posterior distribution of beta and sigma-squared.
There are two sources of uncertainty: uncertainty about the fitted regression mean, and new observation noise around that mean. This is why a prediction interval for an individual future response is wider than an interval for the conditional mean. Under the Normal–Inverse-Gamma model, integrating out beta and sigma-squared gives a Student-t predictive distribution. When

writing its scale, distinguish the parameterisation of the Student-t distribution from its variance.

Board focus: equations from the companion study guide

$$y_* \mid \beta, \sigma^2 \sim \mathcal{N}(x_*^\top \beta, \sigma^2).$$
$$y_* \mid \sigma^2, y \sim \mathcal{N}\left(x_*^\top b_n, \sigma^2(1 + x_*^\top V_n x_*)\right).$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.4 Ridge regression as a Gaussian-prior MAP estimate

Teacher's spoken explanation

Ridge regression can be interpreted as Bayesian regression with a zero-centred normal prior on the coefficients. The normal prior contributes a quadratic penalty on the coefficient magnitudes. When we combine it with the Gaussian likelihood and maximise the posterior, we obtain the ridge estimator.
The amount of shrinkage is controlled by the prior variance. A smaller prior variance means stronger prior precision and stronger pull toward zero. In the conventional objective, the ridge penalty parameter is proportional to the likelihood variance divided by the prior variance. The intercept is often treated separately rather than penalised. The Bayesian interpretation is valuable because it explains the penalty as prior information and, in a full Bayesian analysis, retains uncertainty rather than reporting only the penalised point estimate.

Board focus: equations from the companion study guide

$$\beta \sim \mathcal{N}_p(0, \tau^2 I).$$
$$\exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta) - \frac{1}{2\tau^2}\beta^\top \beta\right\}.$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.5 LASSO as a Laplace-prior MAP estimate

Teacher's spoken explanation

The LASSO penalty is the sum of the absolute values of the coefficients. In Bayesian terms, this corresponds to independent Laplace, or double-exponential, priors centred at zero. The posterior mode minimises the negative log likelihood plus the absolute-value penalty.
The sharp corner of the absolute-value penalty at zero is what allows the MAP estimate to

become exactly zero for some coefficients. In a simple one-dimensional orthogonal case, this gives a soft-thresholding rule: coefficients smaller in magnitude than the threshold are set to zero, while larger coefficients are reduced by the threshold amount. In a multidimensional model with correlated predictors, the solution is not generally obtained by applying this rule independently to each coefficient. Also, for a continuous Laplace prior, the posterior probability that a coefficient is exactly zero remains zero; exact posterior point mass at zero requires a discrete spike component.

Board focus: equations from the companion study guide

$$p(\beta_j) = \frac{\lambda}{2} e^{-\lambda|\beta_j|}.$$

$$\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.6 Spike-and-slab priors and Bayesian variable selection

Teacher's spoken explanation

Spike-and-slab priors introduce an explicit indicator for whether each predictor is included. The spike is a point mass at zero, while the slab is a continuous distribution for nonzero coefficients. The model indicator therefore has a direct posterior inclusion probability.
We can compare inclusion and exclusion through prior odds and a Bayes factor. The posterior inclusion odds equal prior inclusion odds multiplied by the Bayes factor for including the predictor. Unlike a continuous shrinkage prior, spike-and-slab gives genuine posterior probability to a coefficient being exactly zero. When many predictors are available, there can be an enormous number of possible models, so computation may require stochastic search or other approximations. Model averaging can preserve uncertainty about which variables belong in the model.

Board focus: equations from the companion study guide

$$\gamma_j \sim \text{Bernoulli}(q_j), \qquad \beta_j \mid \gamma_j = 0 = 0, \qquad \beta_j \mid \gamma_j = 1 \sim \mathcal{N}(0, \tau_j^2).$$

$$\Pr(\gamma_j = 1 \mid y).$$

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

6.7 Ridge, LASSO, and spike-and-slab compared

Teacher's spoken explanation

Let us compare the three approaches in terms of what their priors are doing. Ridge uses a normal prior and usually shrinks all coefficients continuously toward zero. LASSO uses a Laplace prior; its posterior mode can be sparse because of the sharp corner at zero, but the full continuous posterior does not assign positive probability to exact zero. Spike-and-slab has an explicit point mass at zero and therefore supports posterior inclusion and exclusion probabilities.
These approaches answer related but different questions. Ridge emphasises stable estimation. LASSO combines shrinkage and sparse point estimation. Spike-and-slab directly models uncertainty about variable inclusion. The right choice depends on whether we want stabilisation, a sparse point estimate, or a probability distribution over candidate models.

Pause and check understanding

Ask students to identify what the prior contributes to the negative log posterior. They should be able to say whether the penalty is quadratic, absolute-value, or a discrete inclusion mechanism, and what that implies about the posterior.

Narrative transition: the same model must now serve many regions

Our regression model explains variation among individuals, but the data are collected in several districts. It would be wasteful to treat every district as entirely unrelated, and it would also be unrealistic to force all districts to have exactly the same risk. The team needs a model that learns from each district while allowing information to be shared across districts. This is the central motivation for hierarchical Bayes. It introduces another level of unknown quantities: the group-specific parameters are related through population-level hyperparameters.

Chapter 7

Hierarchical Bayes, empirical Bayes, and profile likelihood

Story opening: learning about the groups and the population

We now arrive at a natural extension of everything we have learned. Each district has its own underlying disease rate or regression effect, but these district-level quantities are not unrelated. They may be viewed as draws from a common population distribution. This creates a hierarchy: observations inform district parameters, district parameters inform population hyperparameters, and the hyperparameters describe the distribution across districts. Partial pooling allows the team to learn about a particular district while borrowing strength from the others. We will distinguish full hierarchical Bayes, which retains uncertainty about the hyperparameters, from empirical Bayes, which estimates them and plugs in the estimate.

Opening words for the instructor

Hierarchical models are useful when observations are grouped and group-level parameters are related. The class should understand the flow of information between levels. We will compare no pooling, complete pooling, partial pooling, full Bayes, and empirical Bayes, and then distinguish marginal likelihood from profile likelihood.

7.1 Why hierarchical models are needed

Teacher's spoken explanation

Suppose several groups each have their own parameter, but we believe the groups are related. Estimating each group independently can be unstable when group sample sizes are small. Forcing every group to have exactly the same parameter can ignore real differences. A hierarchical model provides a middle path: each group has its own parameter, and those parameters come from a shared population distribution.

This adds another level of uncertainty. We have the observation model at the data level, the group-parameter distribution at the next level, and hyperparameters that govern the population distribution. Information can flow between groups through the shared hyperparameters. This is the basic reason hierarchical models can stabilise estimates while still allowing genuine group variation.

Board focus: equations from the companion study guide

$$y_j \mid \theta_j \sim p(y_j \mid \theta_j), \qquad \theta_j \mid \eta \sim p(\theta_j \mid \eta), \qquad \eta \sim p(\eta),$$

$$p(\theta_1, \ldots, \theta_J, \eta \mid y_1, \ldots, y_J) \propto p(\eta) \prod_{j=1}^{J} p(\theta_j \mid \eta) p(y_j \mid \theta_j).$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.2 Exchangeability and partial pooling

Teacher's spoken explanation

Exchangeability at the group level means that if we relabel the groups, the joint distribution of their parameters does not change. It is not the same as saying the group parameters are equal. A shared population distribution provides a common structure without forcing every parameter to the same value.
No pooling estimates each group separately. Complete pooling assumes a common parameter. Partial pooling allows differences but shrinks noisy group estimates toward the population distribution. The degree of shrinkage depends on the balance between each group's sampling variance and the between-group variance. A group with imprecise data is pulled more strongly toward the population mean than a group with precise data.

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.3 The Normal–Normal hierarchical model

Teacher's spoken explanation

The Normal–Normal model makes partial pooling algebraically transparent. We observe a group estimate y-bar-j with sampling variance s-j-squared. The true group effect theta-j is itself normal with population mean mu and between-group variance tau-squared. If we condition on mu and tau-squared, the posterior for theta-j is normal.
Its precision is the sum of sampling precision and population-prior precision. Its mean is a precision-weighted average of the group's observed estimate and the population mean. If the group's sampling variance is large, its own observation receives less weight and shrinkage is stronger. If the between-group variance is large, the groups are allowed to differ substantially and the shrinkage is weak. This gives us a direct algebraic expression for partial pooling.

Board focus: equations from the companion study guide

$$\bar{y}_j \mid \theta_j \sim \mathcal{N}(\theta_j, s_j^2), \qquad \theta_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, \tau^2).$$
$$\theta_j \mid \bar{y}_j, \mu, \tau^2 \sim \mathcal{N}(m_j, v_j),$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.4 Marginal likelihood for hyperparameters

Teacher's spoken explanation

The hyperparameters describe the population distribution, so they must be learned from the collection of groups. One way to do that is to integrate the group-specific parameters out of the joint model. The resulting marginal likelihood describes the observed group data conditional on the hyperparameters.
In the Normal–Normal case, integrating theta-j gives a normal distribution for each observed group mean with variance equal to the sampling variance plus the between-group variance. This calculation is useful because it reduces the dimension of the problem and shows how the group summaries inform the population-level parameters. Do not confuse marginalising a parameter by integration with profiling it out by maximisation. Those are different operations.

Board focus: equations from the companion study guide

$$p(y \mid \eta) = \prod_{j=1}^{J} \int p(y_j \mid \theta_j) p(\theta_j \mid \eta) \, \mathrm{d}\theta_j.$$
$$\bar{y}_j \mid \mu, \tau^2 \sim \mathcal{N}(\mu, s_j^2 + \tau^2),$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.5 Empirical Bayes

Teacher's spoken explanation

Empirical Bayes estimates the hyperparameters from the data and then plugs those estimates into the group-level prior. One common method maximises the marginal likelihood after integrating out the group effects. Another uses moments of the observed group estimates. This is convenient because the data themselves help determine the amount of pooling.
The limitation is that the hyperparameters are treated as if their estimates were known exactly.

That ignores uncertainty about the population distribution, which may be substantial when there are only a few groups. Consequently, empirical-Bayes intervals for group effects can be too narrow. Full hierarchical Bayes addresses this by assigning a prior to the hyperparameters and averaging over their posterior uncertainty.

Board focus: equations from the companion study guide

$$\hat{\eta}_{EB} = \arg\max_{\eta} p(y \mid \eta), \qquad p(y \mid \eta) = \int p(y \mid \theta, \eta) p(\theta \mid \eta)\, \mathrm{d}\theta.$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.6 Profile likelihood: the syllabus connection

Teacher's spoken explanation

Profile likelihood is a likelihood-based method for handling nuisance parameters. Suppose psi is the parameter we care about and lambda is a nuisance parameter. For every fixed value of psi, we maximise the likelihood over lambda. The resulting function of psi is called the profile likelihood. This is different from Bayesian marginalisation. Bayesian marginalisation integrates over lambda using a probability distribution. Profiling selects the best-fitting lambda for each psi, while integration averages over possible lambda values. We will work through the Normal group-summary example: for fixed tau-squared, maximise over mu, substitute the estimate back into the likelihood, and then maximise the profile function over the nonnegative range of tau-squared. The divisor J in the resulting maximum-likelihood expression comes from likelihood maximisation; the familiar J-minus-one divisor belongs to the unbiased sample variance.

Board focus: equations from the companion study guide

$$L_p(\psi) = \max_{\lambda} L(\psi, \lambda).$$

$$p(y \mid \psi) = \int p(y \mid \lambda, \psi) p(\lambda \mid \psi)\, \mathrm{d}\lambda,$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

7.7 Full hierarchical Bayes versus empirical Bayes

Teacher's spoken explanation

The central difference is how uncertainty about the hyperparameters is handled. In full hierarchical Bayes, we assign a hyperprior and calculate the joint posterior. To obtain the posterior for a particular group parameter, we integrate its conditional posterior over the posterior distribution of the hyperparameters. In empirical Bayes, we estimate the hyperparameters and plug in one value.
Both approaches use the observed data. The difference is not whether data are used, but whether hyperparameter uncertainty is propagated. Full Bayes is more complete but can be computationally more demanding. Empirical Bayes can be efficient and useful, but it may understate uncertainty, especially when the number of groups is small.

Board focus: equations from the companion study guide

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y)\, \mathrm{d}\eta.$$

$$p_{EB}(\theta_j \mid y) = p(\theta_j \mid \hat{\eta}_{EB}, y).$$

Pause and check understanding

Ask students to draw the levels of the model and identify which quantities are group-specific and which are shared. Then ask what is integrated out, what is maximised, and what is plugged in. This makes the distinctions between hierarchical Bayes, marginal likelihood, empirical Bayes, and profile likelihood explicit.

Narrative transition: turn the whole course into a single method

At this point, the research team has moved from one observation to a population of groups, from a simple posterior update to simulation, and from one coefficient to a structured regression model. The mathematics may look different from chapter to chapter, but the reasoning has remained stable: specify a probability model, represent uncertainty, condition on evidence, calculate or approximate the posterior, summarise it, and interpret it scientifically. The next chapter collects the formulas and derivation patterns that recur throughout this journey. After that, we will revisit the examination questions as individual episodes from the same investigation.

Chapter 8

Formula atlas, derivation templates, and exam checklist

Story opening: recognising the same reasoning beneath different formulas

Before we enter the question bank, pause and look back. The Beta–Bernoulli update, the Gamma–Poisson update, the Normal–Inverse-Gamma regression posterior, and the conditional distributions used in Gibbs sampling may look like separate calculations. But they all ask us to identify a likelihood, combine it with a prior, and understand the resulting distribution. The formula atlas is not a list to memorise blindly. It is a set of recognisable patterns. In an examination, the student who can identify the model and explain the role of every term is much safer than the student who remembers a formula without understanding its origin.

Opening words for the instructor

This final chapter is a teaching and revision chapter. Use it to rehearse the formulas and the derivation patterns aloud. The objective is not to memorise isolated expressions but to know how to reconstruct them under examination conditions.

8.1 One-page formula atlas

Teacher's spoken explanation

This formula atlas is a map, not a substitute for understanding. When you meet a model, first identify the sampling distribution and the unknown parameter. Then recognise the conjugate prior or posterior expression. The formula tells you the update, but you must still check the support and parameterisation.

A useful way to revise is to hide the right-hand column and reconstruct each result from the likelihood and prior kernels. For MCMC, remember that the Metropolis–Hastings formula includes the reverse proposal divided by the forward proposal. For ridge, the prior precision creates the penalty. For hierarchical normal models, the shrinkage weight depends on both the between-group variance and sampling variance.

Pause and check understanding

Ask students to reconstruct one formula from its likelihood and prior rather than reading it from the page. The final exam answer should show the reasoning, not only the final expression.

8.2 Derivation templates graders expect

Teacher's spoken explanation

A strong examination solution is organised so that every step can be checked. For conjugacy, write the likelihood kernel, write the prior kernel, multiply them, collect powers and exponential terms, identify the family, and state the updated parameters. For a normal posterior, collect the quadratic and linear terms and complete the square.
For Bayes factors, calculate marginal likelihoods by integration, form the ratio, multiply by prior odds, and convert to probabilities if required. For MCMC, identify the target and proposal, write the full acceptance ratio, and explain the diagnostics. For hierarchical models, draw the levels of the model, write the joint posterior, derive the group-level conditional distribution, and explain shrinkage. A grader should be able to follow the logic without guessing what you have omitted.

Pause and check understanding

Ask students to reconstruct one formula from its likelihood and prior rather than reading it from the page. The final exam answer should show the reasoning, not only the final expression.

8.3 Common errors that lose marks

Teacher's spoken explanation

I want you to treat this list as a final warning before the examination. Always state the Gamma parameterisation, because rate and scale are easy to confuse. Include a Jacobian when transforming densities. Do not treat the likelihood as a posterior. Use integrated marginal likelihoods rather than maximised likelihoods when computing ordinary Bayes factors. Define what kind of credible interval you are reporting.
For MCMC, include the Hastings correction when the proposal is asymmetric, and do not declare convergence based on one trace plot or one chain. Keep posterior uncertainty separate from Monte Carlo error. For regression, distinguish continuous shrinkage from a point mass at zero. Finally, state that empirical Bayes plugs in hyperparameter estimates and therefore does not fully propagate their uncertainty.

Pause and check understanding

Ask students to reconstruct one formula from its likelihood and prior rather than reading it from the page. The final exam answer should show the reasoning, not only the final expression.

Chapter 9

Teacher's lecture scripts for all 63 examination questions

The examination is the same investigation, seen through different questions

We have now built the whole Bayesian argument. The questions that follow are not seven unrelated collections. They are seven ways an examiner can ask us to continue the investigation. The first type asks whether we understand evidence and conditional probability. The second asks us to update distributions. The third asks how to summarise the posterior. The fourth asks how to compare hypotheses. The fifth asks how to compute a difficult posterior. The sixth asks how to explain outcomes using predictors. The seventh asks how to connect related groups. As we solve each problem, keep asking: at what stage of the research story are we, and what uncertainty are we trying to describe?

The following section is designed for direct teaching. Every question from the companion study guide appears in its original order. Read the problem aloud, identify the skill being assessed, guide students through the reasoning, and then present the worked steps. The solution is included immediately after the question so the instructor does not need to move between chapters while teaching.

9.1 Type 1 – Bayesian foundations, Bayes' theorem, odds, and prediction

Opening script for this question type

For this set, focus on the direction of conditioning. Every time you see a probability question, define the event and write the information in conditional notation before substituting numbers. In odds questions, use the sequence prior odds, likelihood ratio, posterior odds, and then probability. In predictive questions, remember to average over posterior uncertainty.

9.1.1 Question 1: Fully solved – diagnostic testing

Original question

A condition has prevalence 0.02. A screening test has sensitivity 0.95 and specificity 0.90. Find the posterior probability of disease after a positive result.

Teacher's spoken walkthrough

Start by identifying exactly what is being conditioned on. We know the test is positive, and we want disease probability. I would first define the disease event, then write sensitivity, specificity, and prevalence using their correct conditional directions. The denominator must include every way a positive test can occur. Only after calculating that marginal positive probability should we divide the diseased-positive contribution by it. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

Let D denote disease and $+$ a positive result. Then $\Pr(+ \mid D) = 0.95$ and $\Pr(+ \mid D^c) = 1 - 0.90 = 0.10$. By total probability,

$$\Pr(+) = 0.95(0.02) + 0.10(0.98) = 0.019 + 0.098 = 0.117.$$

Therefore

$$\boxed{\Pr(D \mid +) = \frac{0.95(0.02)}{0.117} = 0.1624.}$$

The result is 16.24%, not 95%. Sensitivity is a conditional probability given disease; the requested quantity conditions on a positive test. Reversing the conditioning is the central error this question is designed to expose.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.2 Question 2: Fully solved – prior and posterior odds

Original question

Two discrete hypotheses have prior probabilities $\Pr(H_1) = 0.30$ and $\Pr(H_0) = 0.70$. Suppose $p(y \mid H_1)/p(y \mid H_0) = 4$. Find posterior odds and posterior probabilities.

Teacher's spoken walkthrough

This is an odds question, so I would deliberately avoid starting with Bayes' probability formula. Convert the two prior probabilities into prior odds. Then use the likelihood ratio to update those odds. At the end, remember that odds are not probabilities: divide the H-one part by the sum of the two odds parts to recover the posterior probability. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

Prior odds are $0.30/0.70 = 3/7$. The likelihood ratio is 4, so posterior odds are $(3/7)4 = 12/7$.

Hence

$$\Pr(H_1 \mid y) = \frac{12}{12+7} = \frac{12}{19} = 0.6316, \quad \Pr(H_0 \mid y) = \frac{7}{19} = 0.3684.$$

The correct sequence is prior odds → multiply by likelihood ratio → posterior odds → convert odds to probabilities.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.3 Question 3: New practice – base-rate calculation

Original question

A rare event has prior probability 0.05. A test has sensitivity 0.90 and false-positive probability 0.08. Compute the posterior probability of the event after a positive result. Show the marginal probability of a positive test before dividing.

Teacher's spoken walkthrough

The key skill is the base-rate calculation. I want you to write the true-positive contribution and the false-positive contribution separately. The marginal positive probability is the sum of those contributions. This forces us not to confuse sensitivity with the posterior probability after a positive result. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Let A be the event and $+$ a positive test. The false-positive probability is 0.08, so

$$\Pr(+) = 0.90(0.05) + 0.08(0.95) = 0.045 + 0.076 = 0.121.$$

Therefore

$$\boxed{\Pr(A \mid +) = \frac{0.045}{0.121} = 0.3719.}$$

The prior prevalence is 5%; conditioning on the positive result raises the probability to about 37.2%. A test's sensitivity cannot be substituted for this posterior probability.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.4 Question 4: New practice – sequential updating

Original question

A coin's success probability has prior $p \sim \text{Beta}(1, 1)$. The sequence observed is H, H, T, H, H. Derive the posterior distribution and the posterior predictive probability of heads on the next toss. Explain why sequential updating gives the same result as updating once with the full sample.

Teacher's spoken walkthrough

This question illustrates sequential learning. Each toss contributes one factor of p for a head or one factor of one-minus-p for a tail. We can update after each observation, or multiply all five likelihood contributions at once. Since multiplication is associative, the final posterior is the same. Count successes and failures, then update the two Beta parameters. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

There are four heads and one tail, so the likelihood is proportional to $p^4(1-p)$. The uniform prior is $\text{Beta}(1, 1)$. Hence

$$p \mid y \sim \text{Beta}(1+4, 1+1) = \boxed{\text{Beta}(5, 2)}.$$

The posterior mean and predictive probability of heads are both $5/(5+2) = 5/7$. Sequential updating gives the same result because each observation multiplies the current posterior kernel by its Bernoulli likelihood; multiplication is associative. The order of the five observations therefore does not affect the final posterior under the iid Bernoulli model.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.5 Question 5: New practice – posterior predictive count

Original question

After Question 4, let Y_* be the number of heads in the next two tosses. Find $\Pr(Y_* = 2 \mid y)$ by integrating over the posterior distribution of p. Do not substitute the posterior mean into a binomial likelihood without justification.

Teacher's spoken walkthrough

Here we move from parameter estimation to prediction. Conditional on p, two heads have probability p-squared. Since p is uncertain, we must average p-squared over its posterior distribution. This is a posterior second moment, not simply the square of the posterior mean. That distinction is exactly what the question is testing. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and

finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Conditional on p, the probability of two heads is p^2. Therefore

$$\Pr(Y_* = 2 \mid y) = \mathbb{E}(p^2 \mid y).$$

For $p \sim \text{Beta}(a, b)$, $\mathbb{E}(p^2) = a(a+1)/[(a+b)(a+b+1)]$. With $(a, b) = (5, 2)$,

$$\boxed{\Pr(Y_* = 2 \mid y) = \frac{5 \cdot 6}{7 \cdot 8} = \frac{15}{28} = 0.5357.}$$

This is not $(\mathbb{E}[p \mid y])^2 = 25/49$ because the predictive calculation averages the conditional probability over posterior uncertainty. The correct expression uses the posterior second moment.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.6 Question 6: Book-aligned – discrete genetic-status inference

Original question

A woman's prior probability of being a carrier of a recessive gene is 1/2. Conditional on being a carrier, each son independently has probability 1/2 of being unaffected; if she is not a carrier, a son is unaffected with probability 1. She has two unaffected sons. Calculate her posterior carrier probability, then update again if a third son is also unaffected. State the conditional-independence assumption.

Teacher's spoken walkthrough

We have a discrete unknown state: carrier or non-carrier. I would make a two-row table containing prior probability and likelihood of the two unaffected sons. Multiply each row to obtain unnormalised posterior weights, then normalise. For the third son, use the current posterior as the new prior. The independence assumption is conditional on carrier status and the stated inheritance model. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The prior probability of carrier status is 1/2. Given carrier status, the probability of two unaffected sons is $(1/2)^2 = 1/4$. Given non-carrier status, it is 1. Thus

$$\Pr(C \mid U_1, U_2) = \frac{(1/4)(1/2)}{(1/4)(1/2) + 1(1/2)} = \frac{1/8}{5/8} = \boxed{1/5}.$$

For a third unaffected son, use the previous posterior as the new prior:

$$\Pr(C \mid U_1, U_2, U_3) = \frac{(1/2)(1/5)}{(1/2)(1/5) + 1(4/5)} = \frac{1/10}{9/10} = \boxed{1/9}.$$

The two sons' outcomes are assumed conditionally independent given the mother's carrier status. If the model allowed mutation or dependence, the likelihoods would need to change.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.7 Question 7: Book-aligned – spelling or classification inference

Original question

A typed string has three possible intended words. Their prior probabilities are proportional to 7.60×10^{-5}, 6.05×10^{-6}, and 3.12×10^{-7}. The likelihoods of the observed string under these possibilities are 0.00193, 0.000143, and 0.975. Compute and interpret all three posterior probabilities after normalisation.

Teacher's spoken walkthrough

There are three possible intended words. For each candidate, multiply its prior weight by the probability of observing the typed string under that candidate. These three products are unnormalised posterior weights. Add them and divide each weight by their total. The priors are stated as proportional values, so their common normalising factor cancels. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Compute unnormalised posterior weights:

$$w_1 = (7.60 \times 10^{-5})(0.00193) = 1.4668 \times 10^{-7},$$

$$w_2 = (6.05 \times 10^{-6})(0.000143) = 8.6515 \times 10^{-10}, \quad w_3 = (3.12 \times 10^{-7})(0.975) = 3.0420 \times 10^{-7}.$$

The sum is $W = 4.5174515 \times 10^{-7}$. Dividing each weight by W gives approximately

$$\boxed{(0.3247,\ 0.0019,\ 0.6734).}$$

The third candidate has the greatest posterior probability, followed by the first; the second has very little posterior support. The conclusion is conditional on the candidate set and the supplied prior and error probabilities. Omitting plausible candidate words would make the reported posterior probabilities conditional on an incomplete set.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.8 Question 8: Book-aligned – likelihood principle and odds

Original question

Explain why the likelihood ratio updates prior odds to posterior odds for two simple hypotheses. For a numerical illustration, use prior odds $2:3$ and likelihood ratio $5:2$. Obtain posterior odds and probabilities.

Teacher's spoken walkthrough

The likelihood principle is visible in the odds identity: data alter the odds by the likelihood ratio. With prior odds two to three, multiply the numerator odds by five and denominator odds by two, or equivalently multiply the odds by five-halves. Then convert the resulting odds to a probability. Explain that evidence changes relative support, not by multiplying a probability directly. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Prior odds are $2/3$ and the likelihood ratio is $5/2$. Posterior odds are

$$\frac{2}{3} \times \frac{5}{2} = \frac{5}{3}.$$

Thus $\Pr(H_1 \mid y) = 5/(5+3) = 5/8$ and $\Pr(H_0 \mid y) = 3/8$. The likelihood principle says that, under the adopted model, the data change posterior odds through the likelihood ratio. It does not say that the likelihood model itself is immune to misspecification.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

9.1.9 Question 9: Book-aligned – model construction

Original question

A dataset consists of observations from multiple exchangeable units. Explain the three steps of Bayesian data analysis and distinguish (i) a parameter, (ii) an observed outcome, (iii) an explanatory variable, and (iv) a future observation. Explain why conditional iid sampling can induce marginal dependence after the parameter is integrated out.

Teacher's spoken walkthrough

This is a model-construction question rather than a numerical one. Begin with the full joint probability model, condition on the observed data, and evaluate model fit and sensitivity. Then define the four types of quantities. To explain marginal dependence, emphasise that a shared unknown parameter connects observations once it is integrated out, even though they are independent conditional on that parameter. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish

with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The three steps are: (i) specify the joint probability model, including the distribution of observations conditional on unknown parameters and the prior; (ii) condition on the observed data to obtain the posterior; and (iii) evaluate model fit, substantive implications, and prior/model sensitivity. A parameter is an unobserved quantity governing the data distribution; an observed outcome is the realised response; an explanatory variable is a covariate used to condition the response model; and a future observation is a not-yet-observed but potentially observable outcome. If Y_i are conditionally iid given a shared θ, their joint conditional density is a product. After integrating over θ, the shared uncertainty induces marginal dependence: observing one unit changes the posterior for θ and hence changes predictions for another unit.

Closing words for the instructor

The conclusion should be stated in context, not only as a decimal. Point out the assumption that makes the calculation valid and name the common error the question was designed to catch. Then ask students what would change if the prior information or likelihood changed.

Connecting question type 1 to type 2

The first set established how evidence changes probabilities. Now we ask how to perform that update repeatedly and efficiently when observations follow familiar sampling distributions. The next questions turn Bayes' rule into posterior calculations for Bernoulli, Binomial, Poisson, Exponential, categorical, and Normal models.

9.2 Type 2 – exponential families and conjugate models

Opening script for this question type

In these questions, the main task is conjugate updating. The reliable method is to write the likelihood kernel and prior kernel separately, multiply them, and identify the posterior family. I will say the update aloud as we write it: prior hyperparameter plus the sufficient statistic contributed by the observations.

9.2.1 Question 1: Fully solved – Beta–Binomial update

Original question

Suppose $p \sim \text{Beta}(2, 3)$ and $Y \mid p \sim \text{Binomial}(10, p)$. Eight successes are observed. Find the posterior distribution, posterior mean, and predictive probability of success on the next trial.

Teacher's spoken walkthrough

Recognise the Binomial likelihood kernel and the Beta prior kernel. The number of successes adds to the first Beta shape and the number of failures adds to the second. Then compute the posterior mean. A future Bernoulli success has probability equal to the posterior expectation of p, so the predictive probability is the same mean. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to

the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The likelihood kernel is $p^8(1-p)^2$ and the prior kernel is $p^{2-1}(1-p)^{3-1}$. Multiplication gives $p^9(1-p)^4$. Matching the Beta kernel,

$$p \mid y \sim \text{Beta}(2+8, 3+10-8) = \boxed{\text{Beta}(10,5)}.$$

The posterior mean is $10/(10+5) = 2/3$. The predictive probability of a success in the next Bernoulli trial is the posterior expectation of p, hence $2/3$.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.2 Question 2: Fully solved – Gamma–Poisson with exposure

Original question

Let counts satisfy $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$. A prior is $\lambda \sim \text{Gamma}(2, 1.5)$ (shape-rate). Three observations have counts $(2,1,4)$ and exposures $(1,2,3)$. Find the posterior distribution and posterior mean.

Teacher's spoken walkthrough

The word exposure is the central clue. The total count updates the Gamma shape, while the sum of exposures updates the Gamma rate. I would calculate those two totals separately before writing the posterior. Then use shape divided by rate for the posterior mean. Do not replace total exposure by the number of observations unless every exposure is one. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The total count is $2+1+4=7$ and total exposure is $1+2+3=6$. The likelihood kernel is $\lambda^7 e^{-6\lambda}$. The prior kernel is $\lambda^{2-1}e^{-1.5\lambda}$. Their product is proportional to $\lambda^8 e^{-7.5\lambda}$, which is a Gamma kernel with shape 9 and rate 7.5:

$$\boxed{\lambda \mid y \sim \text{Gamma}(9, 7.5)}, \qquad \mathbb{E}(\lambda \mid y) = \frac{9}{7.5} = 1.2.$$

The rate update adds exposure, not the number of observations. Confusing the two would incorrectly produce a posterior rate of 4.5 instead of 7.5.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel

from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.3 Question 3: New practice – Gamma–Exponential

Original question

Waiting times are iid Exponential with unknown rate λ. The prior is Gamma$(3, 2)$ (shape-rate). Four waiting times sum to 10. Derive the posterior and calculate its mean and variance.

Teacher's spoken walkthrough

For exponential waiting times, each exact observation adds one event to the shape, and the total time adds to the rate. Combine the prior Gamma kernel with the likelihood kernel. Once the posterior shape and rate are identified, use the Gamma formulas for the mean and variance, keeping in mind that this manual uses shape-rate parameterisation. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

For $n = 4$ exponential observations with total waiting time 10, the likelihood kernel is $\lambda^4 e^{-10\lambda}$. Multiplying by the prior $\lambda^{3-1} e^{-2\lambda}$ gives $\lambda^6 e^{-12\lambda}$, so

$$\lambda \mid y \sim \text{Gamma}(7, 12).$$

The mean is $7/12 = 0.5833$ and the variance is $7/12^2 = 7/144 = 0.04861$. The shape increases by the number of observed events, while the rate increases by total time at risk.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.4 Question 4: New practice – Dirichlet–Multinomial

Original question

A three-category probability vector has prior Dirichlet$(1/2, 1/2, 1/2)$. Observed category counts are $(4, 3, 1)$. Derive the posterior Dirichlet distribution and posterior predictive probabilities for the next category.

Teacher's spoken walkthrough

The Dirichlet update is component-wise. Start with the prior vector of three concentration parameters and add the observed count in each category. The posterior predictive probability for each next category is its updated concentration divided by the sum of all updated concentrations. Check that the three predictive probabilities add to one. As you present the answer, keep the

question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The Dirichlet prior is Dirichlet$(1/2, 1/2, 1/2)$ and the counts are $(4, 3, 1)$. Add counts component-wise:

$$\boldsymbol{p} \mid y \sim \text{Dirichlet}(4.5, 3.5, 1.5).$$

The updated total concentration is $4.5 + 3.5 + 1.5 = 9.5$. Therefore the next-category predictive probabilities are

$$\boxed{\left(\frac{4.5}{9.5}, \frac{3.5}{9.5}, \frac{1.5}{9.5}\right) = (0.4737, 0.3684, 0.1579).}$$

The components sum to one. The posterior mean of each probability equals its predictive probability for the next categorical observation.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.5 Question 5: New practice – Normal–Gamma

Original question

Let $Y_i \mid \mu, \tau \sim \mathcal{N}(\mu, 1/\tau)$, $\mu \mid \tau \sim \mathcal{N}(4, 1/(2\tau))$, and $\tau \sim \text{Gamma}(3, 2)$. A sample has $n = 4$, $\bar{y} = 5$, and $S = \sum_i (y_i - \bar{y})^2 = 10$. Compute $(\kappa_n, \mu_n, a_n, b_n)$ and state the posterior factorisation.

Teacher's spoken walkthrough

This is the Normal–Gamma update. Identify kappa-zero, mu-zero, a-zero, and b-zero from the prior. Then add the sample size to kappa, form the precision-weighted mean, add n over two to the Gamma shape, and update the rate with both the within-sample sum of squares and the prior-data mean discrepancy term. Finally state the conditional Normal posterior for mu and Gamma posterior for precision. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Here $\kappa_0 = 2$, $\mu_0 = 4$, $a_0 = 3$, $b_0 = 2$, $n = 4$, $\bar{y} = 5$, and $S = 10$. Thus

$$\kappa_n = 6, \quad \mu_n = \frac{2(4) + 4(5)}{6} = \frac{28}{6} = 4.6667,$$

$$a_n = 3 + 4/2 = 5,$$

$$b_n = 2 + 10/2 + \frac{2(4)}{2(6)}(5 - 4)^2 = 2 + 5 + \frac{2}{3} = \frac{23}{3} = 7.6667.$$

Hence $\tau \mid y \sim \text{Gamma}(5, 23/3)$ and $\mu \mid \tau, y \sim N(14/3, 1/(6\tau))$. The posterior is a Normal–Gamma distribution.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.6 Question 6: Book-aligned – Poisson rate and exposure

Original question

A sequence of count observations has known exposures x_i and unknown common rate λ. Starting from a Gamma(shape a, rate b) prior, derive the posterior. Identify the sufficient statistics and explain the interpretation of the prior shape and rate contributions.

Teacher's spoken walkthrough

Derive the likelihood as a function of lambda and identify its sufficient statistics. The total number of events is the statistic that updates shape; total exposure updates rate. Explain the prior Gamma hyperparameters in the shape-rate convention and how they contribute to the posterior. The interpretation of pseudo-information should be presented carefully, not as literal additional observed events in every context. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The likelihood kernel is

$$\prod_i (x_i\lambda)^{y_i} e^{-x_i\lambda}/y_i! \propto \lambda^{\sum_i y_i} e^{-\lambda \sum_i x_i}.$$

Multiplying by the Gamma prior kernel $\lambda^{a-1}e^{-b\lambda}$ yields a Gamma posterior with shape $a + \sum_i y_i$ and rate $b + \sum_i x_i$. The sufficient statistics for λ are total count $\sum_i y_i$ and total exposure $\sum_i x_i$. The prior adds $a - 1$ in the power of λ and b in the exponential rate; the posterior mean combines these pseudo-information contributions with observed counts and exposure.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.7 Question 7: Book-aligned – Beta–Binomial predictive distribution

Original question

Starting from a Beta(a, b) prior and a Binomial(n, p) sampling model, derive the marginal probability of observing y successes after integrating out p. Identify the resulting distribution and derive the one-step posterior predictive success probability.

Teacher's spoken walkthrough

To find the Beta–Binomial marginal distribution, integrate the Binomial likelihood against the Beta prior. The binomial coefficient remains outside the integral, and the integral is a ratio of Beta functions. Then derive the predictive success probability from the updated Beta mean. The predictive probability is an average over p, not an independent plug-in calculation. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The marginal probability is

$$p(y) = \binom{n}{y} \int_0^1 p^y (1-p)^{n-y} \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)} \, \mathrm{d}p = \boxed{\binom{n}{y} \frac{B(a+y, b+n-y)}{B(a,b)}}.$$

This is the Beta–Binomial distribution. After observing y successes, $p \mid y \sim \text{Beta}(a+y, b+n-y)$. The next-trial predictive probability is the posterior mean $(a+y)/(a+b+n)$.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.8 Question 8: Book-aligned – Normal mean and variance

Original question

For iid $N(\mu, \sigma^2)$ observations, assume $\sigma^2 \sim \text{InvGamma}(a_0, b_0)$ and $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$. Derive the posterior hyperparameters in terms of n, $\bar{y}$, and $S = \sum (y_i - \bar{y})^2$.

Teacher's spoken walkthrough

For the Normal model with unknown variance, use the Normal–Inverse-Gamma family. First combine the sample mean with the prior mean to obtain the updated location and precision parameter. Then update the shape by n over two and the inverse-Gamma rate with the within-sample variation and the mean-shift term. Keep variance and precision parameterisations separate. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Multiplying the likelihood by the Normal–Inverse-Gamma prior and completing the square gives

$$\kappa_n = \kappa_0 + n, \qquad \mu_n = \frac{\kappa_0 \mu_0 + n\bar{y}}{\kappa_0 + n},$$

$$a_n = a_0 + n/2, \qquad b_n = b_0 + S/2 + \frac{\kappa_0 n}{2(\kappa_0 + n)}(\bar{y} - \mu_0)^2.$$

The last term is the between-centre adjustment: it accounts for disagreement between the prior centre and sample mean. Omitting it would incorrectly treat a sample mean far from the prior mean as if it had no effect on the variance update.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

9.2.9 Question 9: Book-aligned – censoring in an Exponential model

Original question

Suppose $Y_i \mid \lambda \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$. Two exact waiting times equal 1 and 2; a third unit is known only to have survived at least 4 time units. Derive the posterior for λ. Explain why the censored observation adds exposure but not an observed event.

Teacher's spoken walkthrough

Separate exact observations from the censored observation. Each exact time contributes one event and its waiting duration. The unit known to survive at least four time units contributes a survival probability, which adds four units to total exposure but no event. Thus the posterior shape increments by two, while the rate increments by the sum of all observed exact and censored exposure times. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Two exact observations contribute likelihood $\lambda^2 e^{-\lambda(1+2)}$. The right-censored third observation contributes the survival probability $\Pr(Y \geq 4 \mid \lambda) = e^{-4\lambda}$. The total likelihood kernel is therefore $\lambda^2 e^{-7\lambda}$. Multiplying by the prior $\lambda^{a-1} e^{-b\lambda}$ gives

$$\boxed{\lambda \mid \text{data} \sim \text{Gamma}(a + 2, b + 7).}$$

The two exact times contribute two observed failures/events, increasing the shape by two. The censored unit is known to survive through four time units, so it contributes exposure four but no observed event.

Closing words for the instructor

The most important feature here is the pattern of the update. Ask students to identify the sufficient statistic and the exact hyperparameter contribution. If they can reconstruct the kernel from memory and explain the parameterisation, they understand the conjugacy rather than merely recalling a result.

Connecting question type 2 to type 3

We have learned to derive posterior distributions. The next step is to communicate what those distributions say. These questions ask us to choose summaries, distinguish a posterior mean from a MAP estimate, construct credible intervals, and examine how prior choices affect inference.

9.3 Type 3 – posterior summaries, MAP, credible intervals, and priors

Opening script for this question type

These questions ask what we do after the posterior has been derived. We must choose a summary that answers the question: mean, median, MAP, or interval. We will also examine how prior choice and parameter transformation affect the posterior.

9.3.1 Question 1: Fully solved – posterior mean, MAP, and interval

Original question

A binomial sample has $n = 12$ and $y = 8$ successes. The prior is $p \sim \text{Beta}(2, 2)$. Find the posterior, posterior mean, MAP estimate, and central 95% credible interval.

Teacher's spoken walkthrough

The posterior is Beta after adding successes and failures to the prior shapes. We then calculate the posterior mean and, because both shape parameters exceed one, the interior mode. For the central interval, use the 2.5th and 97.5th posterior quantiles. Explain that these are quantiles of the posterior distribution, not a normal approximation unless one is specifically requested. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

Conjugacy gives $p \mid y \sim \text{Beta}(2 + 8, 2 + 4) = \text{Beta}(10, 6)$. Its mean is $10/16 = 0.625$. Since both shape parameters exceed one, its mode is

$$\frac{10 - 1}{10 + 6 - 2} = \frac{9}{14} = 0.6429.$$

The central 95% interval is the pair of posterior quantiles $[q_{0.025}, q_{0.975}]$. Evaluating the Beta CDF numerically gives approximately

$$\boxed{[0.3838,\ 0.8366].}$$

The interval is not formed by adding and subtracting 1.96 standard errors; it is defined by posterior tail probabilities. The mean and MAP differ because the posterior is not exactly symmetric.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.2 Question 2: Fully solved – Jeffreys prior for a binomial probability

Original question

For $Y \sim \text{Binomial}(n, p)$, derive Jeffreys' prior and obtain the posterior after observing $n = 10$ and $y = 7$. Give its posterior mean and explain how it differs from using a uniform prior.

Teacher's spoken walkthrough

First derive Fisher information from the binomial log likelihood. Take the negative expected second derivative, then take its square root to obtain Jeffreys' prior. Combine that Beta one-half, one-half prior with the data. Compare the result to a uniform Beta one, one prior and explain that Jeffreys' construction is motivated by parameterisation invariance. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The log-likelihood, omitting constants, is $\ell(p) = y \log p + (n - y) \log(1 - p)$. Its second derivative is

$$\ell''(p) = -\frac{y}{p^2} - \frac{n - y}{(1 - p)^2}.$$

Taking the negative expectation and using $\mathbb{E}(Y) = np$ gives $I(p) = n/[p(1 - p)]$. Hence Jeffreys' prior is

$$\pi_J(p) \propto [p(1 - p)]^{-1/2}, \quad p \sim \text{Beta}(1/2, 1/2).$$

The posterior is $\text{Beta}(7.5, 3.5)$ and its mean is $7.5/11 = 0.6818$. A uniform prior instead gives $\text{Beta}(8, 4)$ with mean $8/12 = 0.6667$. Jeffreys' prior is not uniform in p; its motivation is invariance under smooth one-to-one reparameterisation.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.3 Question 3: New practice – loss function and Bayes estimator

Original question

Suppose a posterior distribution has mean 4.2, median 3.8, and unique mode 3.1. State the Bayes estimator under squared-error loss, absolute-error loss, and a posterior-mode/MAP criterion. Explain why the answers need not agree.

Teacher's spoken walkthrough

This question connects point estimation to a decision rule. Under squared error, the posterior mean minimises expected posterior loss; under absolute error, a posterior median does. The MAP estimate maximises the posterior density. The same posterior can therefore lead to different Bayes actions because the loss functions are different. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under squared-error loss, posterior expected loss is minimised at the posterior mean, 4.2. Under absolute-error loss, a posterior median minimises expected absolute loss, so use 3.8. The MAP criterion uses the unique mode, 3.1. These need not agree because a posterior distribution can be skewed or multimodal and each loss function values errors differently.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.4 Question 4: New practice – prior sensitivity

Original question

A binomial sample has $n = 10$ and $y = 2$. Compare the posterior means under a Uniform$(0, 1)$ prior and a Beta$(8, 2)$ prior. Explain the direction of the difference and discuss whether either posterior is "data only."

Teacher's spoken walkthrough

A sensitivity analysis means holding the likelihood and data fixed while changing plausible prior specifications. Compare posterior summaries, intervals, and probabilities of substantive events. The conclusion is robust if the substantive interpretation stays similar. Do not choose a prior after looking for one that gives a preferred numerical result. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under the Uniform$(0, 1)$ = Beta$(1, 1)$ prior, the posterior is Beta$(1 + 2, 1 + 8)$ = Beta$(3, 9)$ and its mean is $3/12 = 0.25$. Under Beta$(8, 2)$, the posterior is Beta$(10, 10)$ and its mean is $10/20 = 0.50$. The informative prior puts substantial mass at larger success probabilities and pulls the posterior upward. Neither posterior is "data only": each combines likelihood and prior, though the sample likelihood is identical.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able

to justify the chosen summary in a sentence.

9.3.5 Question 5: New practice – central versus HPD interval

Original question

A posterior density is strongly right-skewed. Explain how to construct a central 90% interval and a 90% HPD region. State two reasons why they can differ and explain why an HPD interval is not generally invariant under a nonlinear transformation.

Teacher's spoken walkthrough

The central interval is built from equal posterior tail probabilities. The HPD region collects the points with greatest density until the required posterior mass is reached. For a skewed or multimodal distribution these need not agree. The key is to explain which property each interval prioritises, not to claim that one is universally superior. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

A central 90% interval is $[q_{0.05}, q_{0.95}]$, leaving exactly 5% posterior mass in each tail. An HPD region is the set of highest-density points containing 90% posterior probability. They differ when the density is skewed (equal tail allocation need not be density-optimal) or multimodal (the HPD set can be disconnected while a central interval bridges low-density gaps). Under a nonlinear transformation, the density is multiplied by a Jacobian, which can reorder which points have highest density; hence HPD sets generally change with parameterisation.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.6 Question 6: Book-aligned – Jeffreys versus uniform prior

Original question

For binomial data with $n = 16$ and $y = 4$, derive the posterior under Jeffreys' prior and under a uniform prior. Compare the posterior means and state when the improper Haldane prior $\text{Beta}(0, 0)$ would fail to give a proper posterior.

Teacher's spoken walkthrough

Compare the two priors by deriving their posterior parameters explicitly. Jeffreys' prior adds one-half to both success and failure counts, while the uniform prior adds one to each. Check propriety at boundary data values for the Haldane prior: if every trial has the same outcome, the corresponding posterior can be improper. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Jeffreys' prior for a binomial probability is Beta$(1/2, 1/2)$. With $n = 16, y = 4$, the posterior is Beta$(4.5, 12.5)$ and mean $4.5/17 = 0.2647$. The uniform prior produces Beta$(5, 13)$ with mean $5/18 = 0.2778$. The Haldane prior has kernel $1/[p(1-p)]$ and corresponds formally to Beta$(0, 0)$. The posterior kernel is $p^{y-1}(1-p)^{n-y-1}$; it is integrable only when $y > 0$ and $n - y > 0$. If all observations are failures or all are successes, the posterior is improper.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.7 Question 7: Book-aligned – normal mean with a flat prior

Original question

Suppose $Y_i \mid \mu \sim N(\mu, \sigma^2)$ with known $\sigma = 2$, sample size 16, and sample mean 3.2. Under the flat prior $\pi(\mu) \propto 1$, derive the posterior and a central 95% credible interval.

Teacher's spoken walkthrough

With a flat prior on mu, the posterior is proportional to the normal likelihood. Complete the square in mu to show that the posterior mean is the sample mean and the posterior variance is sigma-squared divided by n. State that the prior is improper but the posterior is proper under the usual positive sample-size conditions. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

With a flat prior for μ, the posterior is proportional to the normal likelihood. Completing the square gives $\mu \mid y \sim N(\bar{y}, \sigma^2/n) = N(3.2, 4/16) = N(3.2, 0.25)$. The posterior standard deviation is 0.5. A central 95% interval is

$$3.2 \pm 1.96(0.5) = [2.22, 4.18].$$

The interval describes uncertainty about the population mean, not a range expected to contain 95% of individual future measurements.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.8 Question 8: Book-aligned – prior as equivalent information

Original question

For the Beta–Binomial model, explain the pseudo-count interpretation of a Beta(a, b) prior. Use a numerical example to show how the prior and data combine in the posterior mean. State one limitation of taking pseudo-counts literally.

Teacher's spoken walkthrough

The Beta prior contributes a and b to the posterior-mean expression, which motivates the pseudo-count interpretation. Derive the posterior mean and compare its numerator and denominator to the observed successes and total trials. Explain the limitation: a pseudo-count is an algebraic interpretation of the prior update, not necessarily a literal set of observations. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

If $p \sim \text{Beta}(a, b)$ and y successes are observed in n trials, the posterior mean is $(a+y)/(a+b+n)$. The prior contributes a success-like and b failure-like units to this formula, which motivates the pseudo-count interpretation. For example, Beta$(2, 3)$ combined with 7 successes in 10 trials gives Beta$(9, 6)$ and posterior mean $9/15 = 0.6$. The limitation is that prior pseudo-counts summarise a distribution and need not correspond to literal observations or independent sample units; they also do not describe all aspects of prior uncertainty by themselves.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

9.3.9 Question 9: Book-aligned – transformation and propriety

Original question

Let $\eta = \log\{p/(1-p)\}$ be the logit of a binomial probability. Find the prior density on p induced by a flat prior on η. Compare it with a uniform prior on p, and explain the need to check posterior propriety under improper priors.

Teacher's spoken walkthrough

The transformation must be done at the density level. Write eta as logit p, invert to get p as a function of eta, and compute the absolute Jacobian. A flat density on eta becomes proportional to one over p times one-minus-p on the p scale. Then check whether the induced posterior kernel integrates to a finite value for the observed boundary or interior data. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The inverse logit is $p = e^{\eta}/(1+e^{\eta})$, with Jacobian $\mathrm{d}\eta/\,\mathrm{d}p = 1/[p(1-p)]$. A flat prior on η therefore induces

$$\boxed{\pi(p) \propto \frac{1}{p(1-p)}, \quad 0 < p < 1.}$$

This is an improper Beta$(0,0)$ kernel, not a uniform prior on p. The uniform prior on p instead induces a logistic-density prior on η proportional to $p(1-p)$. Under an improper prior, one must verify that the likelihood times prior has a finite integral; otherwise a posterior probability distribution cannot be normalised.

Closing words for the instructor

Finish by explaining what the numerical summary means and what it does not mean. Ask whether a different loss function or prior would change the answer. The student should be able to justify the chosen summary in a sentence.

Connecting question type 3 to type 4

A posterior summary describes an unknown quantity. But many scientific decisions compare explanations rather than estimate one number. We now move from parameter estimation to model comparison, using posterior odds and Bayes factors.

9.4 Type 4 – Bayesian hypothesis testing and Bayes factors

Opening script for this question type

For each test, identify the two models and decide whether each is simple or composite. Calculate marginal likelihoods, form the Bayes factor in the stated direction, and then combine it with prior odds. Keep the model labels visible throughout the calculation.

9.4.1 Question 1: Fully solved – simple binomial hypotheses

Original question

Consider $H_0 : p = 0.5$ and $H_1 : p = 0.7$. Prior probabilities are 0.6 and 0.4 respectively. In 10 Bernoulli trials, 8 successes are observed. Compute BF_{10} and posterior probabilities.

Teacher's spoken walkthrough

Under the null, evaluate the Binomial probability at the fixed value. Under the alternative, integrate the likelihood over the Beta prior. The ratio of these two marginal probabilities is the Bayes factor. Multiply by prior odds to obtain posterior odds, and convert to posterior probability if asked. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

For fixed p, the binomial coefficient cancels in the likelihood ratio. Thus

$$BF_{10} = \frac{0.7^8(0.3)^2}{0.5^8(0.5)^2} = 1.4^8(0.6)^2 \approx 5.3128.$$

The prior odds $H_1 : H_0$ are $0.4/0.6 = 2/3$. Posterior odds are $(2/3)(5.3128) = 3.5419$. Hence

$$\boxed{\Pr(H_1 \mid y) = \frac{3.5419}{1 + 3.5419} = 0.7798, \quad \Pr(H_0 \mid y) = 0.2202.}$$

The Bayes factor is not itself the posterior probability; prior model odds must still be included.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.2 Question 2: Fully solved – normal point null versus a normal alternative

Original question

Assume $Y_i \mid \mu \sim N(\mu, 1)$, with $n = 4$ and $\bar{y} = 0.8$. Compare $H_0 : \mu = 0$ with $H_1 : \mu \sim N(0, 1)$. Compute BF_{10} using the marginal distribution of $\bar{Y}$.

Teacher's spoken walkthrough

For the point null, the marginal likelihood is simply the normal density evaluated at the null mean. Under the alternative, integrate the likelihood over the alternative normal prior. Since both are normal, the integral can be evaluated using the marginal distribution of the observation. Keep the numerator and denominator labels consistent throughout. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

Under H_0, $\bar{Y} \sim N(0, 1/4)$. Under H_1, integrate over the normal prior, giving $\bar{Y} \sim N(0, 1+1/4) = N(0, 1.25)$. Therefore

$$BF_{10} = \frac{\phi(0.8; 0, 1.25)}{\phi(0.8; 0, 0.25)} = \sqrt{\frac{0.25}{1.25}} \exp\left\{-\frac{0.8^2}{2(1.25)} + \frac{0.8^2}{2(0.25)}\right\} \approx 1.2452.$$

The data provide only modest evidence for the alternative. A diffuse alternative prior would spread predictive mass over a wider range and may reduce the marginal density at the observed value.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.3 Question 3: New practice – composite Beta alternative

Original question

Compare $H_0 : p = 0.5$ with $H_1 : p \sim \text{Beta}(2, 2)$. A sample has $n = 10$ and $y = 8$. Derive the marginal likelihood under each model and calculate BF_{10}.

Teacher's spoken walkthrough

This is a composite-alternative problem. The alternative marginal likelihood is an average of the Binomial likelihood over the Beta prior. Write the Beta integral and express it using the Beta function. Then form the Bayes factor against the fixed null value. The prior under the alternative matters because it determines which parameter values receive weight. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under H_0, $p(y \mid H_0) = \binom{10}{8}(0.5)^{10}$. Under H_1, integrate over the Beta$(2, 2)$ prior:

$$p(y \mid H_1) = \binom{10}{8} \frac{B(2 + 8, 2 + 2)}{B(2, 2)}.$$

The combinatorial term cancels in the ratio:

$$BF_{10} = \frac{B(10, 4)}{B(2, 2)(0.5)^{10}} \approx \boxed{2.1483}.$$

The composite alternative has averaged over all values of p using its Beta prior. It is not evaluated only at its MLE $\hat{p} = 0.8$.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.4 Question 4: New practice – prior odds and posterior odds

Original question

A comparison has prior odds $H_1 : H_0 = 1 : 2$. The Bayes factor is $BF_{10} = 1/12$. Compute posterior odds and posterior probabilities. Interpret the evidence direction.

Teacher's spoken walkthrough

The question gives prior odds and a Bayes factor. Do not perform a new likelihood calculation unless asked. Multiply the two to get posterior odds, then convert. State clearly which hypothesis is in the numerator so that a Bayes factor below one is interpreted correctly. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an

unexplained formula.

Worked reasoning and final answer

Prior odds $H_1 : H_0$ are $1/2$. Multiplication by $BF_{10} = 1/12$ gives posterior odds $1/24$. Thus

$$\Pr(H_1 \mid y) = \frac{1}{25} = 0.04, \qquad \Pr(H_0 \mid y) = \frac{24}{25} = 0.96.$$

The Bayes factor favours H_0 by a factor of 12, and the prior already favoured H_0 by a factor of 2. The posterior consequently favours H_0 by a factor of 24.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.5 Question 5: New practice – unequal prior model probabilities

Original question

Two models have prior probabilities 0.3 and 0.7. Their marginal likelihoods are 0.02 and 0.01. Calculate the posterior model probability of the first model. Explain why comparing the marginal likelihoods alone is insufficient when prior model probabilities differ.

Teacher's spoken walkthrough

Each model receives a prior probability. Multiply that prior model probability by the model's marginal likelihood to get its unnormalised posterior weight. Add the weights and normalise. This is the discrete-model version of Bayes' rule and makes it clear that models can have unequal prior probabilities. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Posterior model weights are prior probability times marginal likelihood. The unnormalised weights are $0.3(0.02) = 0.006$ and $0.7(0.01) = 0.007$. The first model's posterior probability is

$$\boxed{\frac{0.006}{0.006 + 0.007} = \frac{6}{13} = 0.4615.}$$

Although its marginal likelihood is twice that of the second model, its prior probability is less than half as large, so its posterior probability is below one half.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.6 Question 6: Book-aligned – Bayes factor from a Beta–Binomial model

Original question

Compare $H_0 : p = 0.5$ with $H_1 : p \sim \text{Beta}(1, 1)$. Four Bernoulli trials produce three successes. Calculate the marginal likelihood under each hypothesis and the Bayes factor. If prior model probabilities are equal, find $\Pr(H_1 \mid y)$.

Teacher's spoken walkthrough

For a Beta–Binomial alternative, integrate the Binomial likelihood over p using the Beta prior. Under the point null, evaluate the likelihood at the fixed p. The Bayes factor is the ratio of the two. This is a common conjugate calculation, and retaining the Binomial coefficient in both marginal likelihoods prevents accidental loss of constants. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under $H_0 : p = 1/2$, the marginal likelihood is $\binom{4}{3}(1/2)^4$. Under H_1, integrating the uniform Beta$(1, 1)$ prior yields

$$p(y \mid H_1) = \binom{4}{3} \frac{B(4, 2)}{B(1, 1)}.$$

Since $B(4, 2) = 1/20$ and $B(1, 1) = 1$, while $(1/2)^4 = 1/16$,

$$BF_{10} = \frac{1/20}{1/16} = \boxed{0.8}.$$

With equal prior model probabilities, posterior odds are 0.8:1, so $\Pr(H_1 \mid y) = 0.8/1.8 = \boxed{4/9 = 0.4444}$. Even with three successes in four trials, the integrated alternative is not strongly supported relative to the point null because it averages over a range of p values.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.7 Question 7: Book-aligned – improper prior and model comparison

Original question

Explain why an improper prior can sometimes produce a proper posterior for parameter estimation but cannot generally be used to calculate an ordinary Bayes factor. Identify the arbitrary constant that causes the problem.

Teacher's spoken walkthrough

An improper prior does not have a finite normalising constant. That can sometimes be harmless when deriving a posterior for a parameter within one model, because the constant cancels. In Bayes factors, however, different model-specific arbitrary constants generally do not cancel.

Therefore an ordinary Bayes factor cannot be trusted with such an improper prior without special construction. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

An improper prior is specified only up to a multiplicative constant: $\pi(\theta) = c\,g(\theta)$, but no value of c makes it integrate to one. In posterior estimation, this constant may cancel between numerator and denominator when the posterior is proper. In a marginal likelihood, however, $p(y \mid H) = \int p(y \mid \theta, H) c g(\theta)\,\mathrm{d}\theta$ retains the arbitrary constant c. If different models have different arbitrary constants, their ratio is arbitrary. Therefore the ordinary Bayes factor is not defined unless the relevant parameter priors are proper or a special construction is provided.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.8 Question 8: Book-aligned – posterior odds under sequential evidence

Original question

Two independent datasets produce Bayes factors $BF_{10}^{(1)} = 2.5$ and $BF_{10}^{(2)} = 4$. Prior odds are $1:3$. Compute the combined posterior odds and posterior model probabilities. State the conditional independence assumption required for multiplication.

Teacher's spoken walkthrough

If the two evidence contributions are conditionally independent under each fixed hypothesis, the joint likelihood ratio is the product of the individual likelihood ratios. Multiply the prior odds by that product. Explain that this sequential updating is equivalent to updating once using the joint data. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under the stated conditional independence and fixed hypotheses, the joint likelihood ratio is the product of the two likelihood ratios, so $BF_{10} = 2.5(4) = 10$. Prior odds are $1/3$, hence posterior odds are $10/3$. Converting to probability gives

$$\Pr(H_1 \mid y_1, y_2) = \frac{10}{13} = 0.7692, \quad \Pr(H_0 \mid y_1, y_2) = \frac{3}{13} = 0.2308.$$

The product rule requires the datasets to be conditionally independent given each hypothesis, or more generally that the supplied Bayes factors be sequentially valid conditional Bayes factors.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

9.4.9 Question 9: Book-aligned – diffuse priors and the Jeffreys–Lindley phenomenon

Original question

Explain qualitatively how a very diffuse prior under a continuous alternative can reduce its marginal likelihood even when the maximum likelihood under that alternative is much larger than under a point null. Why can a small classical p-value coexist with a Bayes factor that favours the null?

Teacher's spoken walkthrough

The alternative marginal likelihood averages over the prior, so a very diffuse prior spreads probability over many parameter values that may predict the observed data poorly. This can reduce the marginal likelihood even if the alternative has a high maximised likelihood. A p-value and a Bayes factor compare different objects, so they can point in different directions. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Under a composite alternative, the marginal likelihood is an average of the likelihood across the prior: $p(y \mid H_1) = \int p(y \mid \theta, H_1)\pi(\theta \mid H_1)\,\mathrm{d}\theta$. A very diffuse prior assigns substantial probability to parameter values that predict data unlike the observation, reducing the average even if the likelihood is sharply maximised at a value that fits the data well. A classical p-value measures tail extremeness under the null sampling distribution; a Bayes factor compares marginal likelihoods under two fully specified models. They answer different questions, so a small p-value and a Bayes factor favouring the null can coexist. The phenomenon is commonly called the Jeffreys–Lindley paradox and is sensitive to the alternative prior scale.

Closing words for the instructor

Finish by saying which hypothesis is favoured and by how much the data changed the prior odds. Remind the class that the result is conditional on the specified priors and models, and that the Bayes factor is not itself a posterior probability.

Connecting question type 4 to type 5

Bayes factors and posterior distributions may require integrals that cannot be evaluated analytically. The next questions ask how to generate representative draws from a posterior and how to decide whether the simulation is reliable.

9.5 Type 5 – Gibbs sampling, Metropolis–Hastings, and convergence diagnostics

Opening script for this question type

These questions examine whether you can construct and assess a Markov chain. Derive full conditionals for Gibbs sampling, include the proposal correction for Metropolis–Hastings, and interpret graphical diagnostics. A long chain is not automatically a trustworthy chain.

9.5.1 Question 1: Fully solved – Gibbs sampler for a correlated bivariate normal

Original question

Consider the target density of a standard bivariate normal with correlation ρ:

$$\pi(x, y) \propto \exp\left[-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right].$$

Derive the two full conditional distributions and state one Gibbs sweep.

Teacher's spoken walkthrough

For a correlated bivariate normal target, each full conditional is a univariate normal distribution. Derive the conditional mean and variance by holding the other variable fixed and completing the square. Then alternate the two conditional draws. The key lesson is that Gibbs sampling constructs a joint chain from accessible conditional distributions. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

Collect terms involving x while holding y fixed:

$$x^2 - 2\rho xy = (x - \rho y)^2 - \rho^2 y^2.$$

Thus the conditional kernel in x is $\exp\{-(x-\rho y)^2/[2(1-\rho^2)]\}$, giving

$$X \mid Y = y \sim N(\rho y, 1 - \rho^2).$$

By symmetry, $Y \mid X = x \sim N(\rho x, 1 - \rho^2)$. A systematic Gibbs sweep from $(x^{(t)}, y^{(t)})$ is

$$x^{(t+1)} \sim N(\rho y^{(t)}, 1 - \rho^2), \qquad y^{(t+1)} \sim N(\rho x^{(t+1)}, 1 - \rho^2).$$

Notice that the second update uses the newly drawn $x^{(t+1)}$, not the old $x^{(t)}$. If $|\rho|$ is close to one, the conditional variance is small and the chain can move slowly along the elongated joint posterior.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.2 Question 2: Fully solved – Metropolis–Hastings acceptance

Original question

The target is standard normal, $\pi(x) \propto e^{-x^2/2}$. A symmetric random-walk proposal moves from current $x = 0$ to candidate $x^* = 1.5$. Find the acceptance probability. State the decision if $U = 0.20$ and if $U = 0.70$.

Teacher's spoken walkthrough

Start by writing the Metropolis–Hastings ratio, including the target density and forward and reverse proposal densities. Because the proposal is symmetric in this question, the proposal terms cancel. Calculate the target density ratio and compare a uniform random number to the acceptance probability. If rejected, the chain stays at its current point. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

For a symmetric proposal, the proposal terms cancel:

$$\alpha = \min\left\{1, \frac{\pi(1.5)}{\pi(0)}\right\} = \min\{1, e^{-1.5^2/2}\} = e^{-1.125} \approx 0.3247.$$

If $U = 0.20$, then $U < \alpha$ and the move is accepted. If $U = 0.70$, the move is rejected and the chain remains at 0. A rejected proposal is still an MCMC iteration and the repeated state must be retained.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.3 Question 3: New practice – full conditionals in a normal hierarchy

Original question

Let $Y_i \mid \theta \sim N(\theta, 1)$ for $i = 1, 2, 3$, with $\bar{y} = 2$. Let $\theta \mid \mu \sim N(\mu, 1)$ and $\mu \sim N(0, 10)$. Derive the full conditional distributions $\theta \mid \mu, y$ and $\mu \mid \theta, y$ and write the Gibbs update order.

Teacher's spoken walkthrough

The two levels form a Normal–Normal hierarchy. For theta given mu, combine the three normal observations with the unit-variance prior. For mu given theta, combine the normal prior for mu with the conditional model for theta. State the conditional distributions and then specify the update order. Each block is sampled using the current values of the other blocks. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The data likelihood for θ is $\prod_{i=1}^{3}\phi(y_i-\theta)$, whose kernel as a function of θ is $\exp\{-3(\theta-\bar{y})^2/2\}$. The prior $\theta \mid \mu \sim N(\mu,1)$ contributes $\exp\{-(\theta-\mu)^2/2\}$. Thus

$$\theta \mid \mu, y \sim N\left(\frac{3\bar{y}+\mu}{4}, \frac{1}{4}\right) = N\left(\frac{6+\mu}{4}, \frac{1}{4}\right).$$

For $\mu \mid \theta, y$, the data are conditionally irrelevant once θ is known. The prior $\mu \sim N(0,10)$ and likelihood $\theta \mid \mu \sim N(\mu,1)$ give posterior precision $1+1/10 = 11/10$ and mean $(\theta/1 + 0/10)/(11/10) = 10\theta/11$. Hence

$$\mu \mid \theta, y \sim N\left(\frac{10\theta}{11}, \frac{10}{11}\right).$$

A Gibbs sweep first draws $\theta^{(t+1)}$ conditional on $\mu^{(t)}$ and the data, then draws $\mu^{(t+1)}$ conditional on the new $\theta^{(t+1)}$.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.4 Question 4: New practice – Metropolis ratio for a Beta target

Original question

The target density is $\pi(p) \propto p(1-p)^2$ on $(0,1)$, corresponding to a Beta$(2,3)$ distribution. A symmetric proposal moves from $p = 0.3$ to $p^* = 0.6$. Compute the acceptance probability. State whether the proposal is accepted when $U = 0.5$.

Teacher's spoken walkthrough

Since the proposal is symmetric, the acceptance probability is the minimum of one and the ratio of the target at the proposed point to the target at the current point. Evaluate the Beta kernel at 0.6 and 0.3. Then compare the result with U equals 0.5. Do not forget that the proposal is accepted when U is no larger than the acceptance probability. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

For a symmetric proposal, the acceptance ratio is the target density ratio:

$$r = \frac{0.6(0.4)^2}{0.3(0.7)^2} = \frac{0.096}{0.147} = 0.65306.$$

Therefore $\alpha = 0.65306$. Since $U = 0.5 < \alpha$, accept the proposal and set the next state to 0.6. If the draw exceeded 0.65306, the chain would stay at 0.3.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.5 Question 5: New practice – interpret graphical diagnostics

Original question

Three trace plots are described: (A) all chains fluctuate around the same level with no visible drift; (B) each chain looks stable alone, but the chains occupy visibly different levels; (C) a chain slowly drifts upward throughout the run. For each, state whether the diagnostic is reassuring and what further action is appropriate.

Teacher's spoken walkthrough

For each trace plot, I want you to separate three questions: do the chains look stationary, do different chains overlap, and do they move enough to explore the target? Plot A is reassuring but should still be supported by other diagnostics. Plot B suggests different chains may be trapped in different regions. Plot C shows drift and is not suitable for posterior summaries. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Plot A is reassuring but should be supplemented by marginal-density and autocorrelation checks and assessed for all important estimands. Plot B is not reassuring: stable-looking individual chains can still target different parts of the space. Use longer runs, better initialisation or a different algorithm, and examine whether a multimodal target or poor mixing is responsible. Plot C shows nonstationary drift; do not summarise it as if it were a sample from the posterior. Reparameterise or improve the sampler, extend warm-up only when justified, and rerun diagnostics.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.6 Question 6: Book-aligned – multiple dispersed chains

Original question

Explain why MCMC should be run from multiple overdispersed starting values. Distinguish stationarity from mixing. Describe what a trace plot, overlaid marginal-density plot, and autocorrelation plot would reveal.

Teacher's spoken walkthrough

Multiple dispersed starts reduce the risk that every chain gives the same misleading picture because they all began in one region. Stationarity means the distribution of the chain has stabilised; mixing means the chain explores that distribution efficiently. Trace, marginal-density and autocorrelation plots answer related but different diagnostic questions. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Dispersed initial values make it harder for chains to give a false impression of convergence merely because they all began in the same small region. If chains converge and mix, between-chain variation should become comparable to within-chain variation and their traces should overlap in the target region. Stationarity means the distribution of the chain no longer changes with iteration; mixing concerns how effectively it moves through the target distribution. Trace plots show time ordering, trends, and chain separation; marginal-density overlays show distributional differences across chains; autocorrelation plots show dependence between successive draws. A good analysis examines all three rather than relying on one summary.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.7 Question 7: Book-aligned – effective sample size

Original question

A single retained chain has $N = 10,000$ draws and approximately geometric autocorrelation $\rho_t = 0.5^t$. Derive the integrated autocorrelation factor and calculate the approximate effective sample size.

Teacher's spoken walkthrough

The geometric autocorrelation sequence can be summed using the infinite geometric series. The integrated autocorrelation factor is one plus twice the sum of positive-lag autocorrelations. Divide the raw draw count by that factor to obtain the approximate effective sample size. Be careful about whether the lag index starts at zero or one; rho-zero is one and is not included in the positive-lag sum. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

For $\rho_t = 0.5^t$,

$$\sum_{t=1}^{\infty} \rho_t = \frac{0.5}{1 - 0.5} = 1.$$

The integrated autocorrelation factor is $1 + 2(1) = 3$. Therefore

$$\boxed{n_{\text{eff}} \approx 10,000/3 = 3,333.3.}$$

The chain contains 10,000 recorded states but, for estimating a posterior mean under this autocorrelation approximation, it carries information comparable to roughly 3,333 independent draws.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.8 Question 8: Book-aligned – Monte Carlo standard error

Original question

A posterior standard deviation is estimated as 2 and effective sample size is 400. Approximate the Monte Carlo standard error of the posterior mean. Explain why this is not a 95% credible-interval half-width.

Teacher's spoken walkthrough

Monte Carlo standard error for a posterior mean is approximately posterior standard deviation divided by the square root of effective sample size. This measures numerical uncertainty in our estimate of the posterior mean due to finite simulation. It is not the posterior uncertainty about the parameter itself, so do not confuse it with a credible interval width. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The Monte Carlo standard error is approximately posterior SD divided by the square root of effective sample size:

$$\text{MCSE} = \frac{2}{\sqrt{400}} = \boxed{0.1}.$$

A 95% credible interval describes posterior uncertainty in the parameter and would depend on posterior quantiles. The MCSE describes the numerical uncertainty in estimating a posterior mean from a finite correlated simulation. It is not the posterior standard deviation and not a credible-interval half-width.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

9.5.9 Question 9: Book-aligned – convergence and dependent draws

Original question

A researcher reports 100,000 MCMC iterations from one chain, discards the first 1,000 draws, and gives posterior means. List at least five missing checks or reporting details needed before accepting the results. Explain why a large raw draw count does not guarantee high information content.

Teacher's spoken walkthrough

A raw iteration count is not a convergence diagnostic. I would ask for multiple chains, dispersed initial values, trace plots, marginal-density comparisons, autocorrelation plots, effective sample sizes, Monte Carlo errors, warm-up justification, and a clear description of the algorithm. Strong autocorrelation means many saved draws can contain little independent information. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

At minimum, report: independent multiple chains; dispersed initial values; overlaid trace plots; assessment of trends and stationarity; marginal-density comparisons across chains; autocorrelation plots; effective sample sizes for every key parameter and derived quantity; Monte Carlo standard errors; a justified warm-up decision; and a description of the transition algorithm and proposal tuning or Gibbs full conditionals. A raw count does not account for autocorrelation: if adjacent draws are highly dependent, 100,000 draws may correspond to far fewer independent draws. A single chain can also remain trapped in one region and never reveal that another posterior mode exists.

Closing words for the instructor

Close by distinguishing the target distribution from the algorithm used to explore it. Ask students what diagnostic evidence supports the result and what warning sign would make them rerun or reparameterise the sampler.

Connecting question type 5 to type 6

Simulation gives us a way to work with complicated posteriors. We now apply Bayesian reasoning to a model with many predictors. The questions connect the normal likelihood and conjugate prior to ridge shrinkage, LASSO, and variable selection.

9.6 Type 6 – Bayesian linear models and regularisation

Opening script for this question type

These problems connect probability models to regression penalties. Begin with the likelihood and prior, derive the posterior or posterior mode, and interpret the regularisation. Be especially careful about the distinction between continuous shrinkage and genuine posterior probability of exclusion.

9.6.1 Question 1: Fully solved – Normal–Inverse-Gamma regression update

Original question

For $y \mid \beta, \sigma^2 \sim N_n(X\beta, \sigma^2 I)$, take prior $\beta \mid \sigma^2 \sim N_p(b_0, \sigma^2 V_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, d_0)$. Derive the posterior parameters.

Teacher's spoken walkthrough

Begin with the normal likelihood and the conditional normal prior for beta. Collect the quadratic terms in beta to identify the posterior precision matrix. Complete the square to obtain the posterior mean. Then identify the remaining inverse-Gamma kernel for sigma-squared. State both conditional posterior distributions and the updated shape and rate. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The likelihood times the conditional normal prior has exponent

$$-\frac{1}{2\sigma^2}\left[(y - X\beta)^\top (y - X\beta) + (\beta - b_0)^\top V_0^{-1}(\beta - b_0)\right].$$

Collect the quadratic and linear terms in β. The posterior precision matrix is $V_0^{-1} + X^\top X$, so

$$V_n = (V_0^{-1} + X^\top X)^{-1}, \qquad b_n = V_n(V_0^{-1} b_0 + X^\top y).$$

Completing the square gives $(\beta - b_n)^\top V_n^{-1}(\beta - b_n)$ plus a term independent of β. The remaining inverse-gamma kernel has shape increased by $n/2$ and rate increased by half the residual-plus-prior quadratic form:

$$a_n = a_0 + n/2, \quad d_n = d_0 + \tfrac{1}{2}(y^\top y + b_0^\top V_0^{-1} b_0 - b_n^\top V_n^{-1} b_n).$$

Therefore $\beta \mid \sigma^2, y \sim N(b_n, \sigma^2 V_n)$ and $\sigma^2 \mid y \sim \text{InvGamma}(a_n, d_n)$.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.2 Question 2: Fully solved – Ridge as Gaussian-prior MAP

Original question

Suppose $X^\top X = \text{diag}(4, 9)$, $X^\top y = (8, 18)^\top$, the likelihood variance is $\sigma^2 = 4$, and the prior is $\beta \sim N(0, 4I)$. Find the ridge/MAP estimate.

Teacher's spoken walkthrough

The Gaussian prior has covariance four times the identity, and the likelihood variance is four. The ridge penalty parameter is their ratio, so it equals one. Add the identity matrix to X-transpose-X, invert the resulting diagonal matrix, and multiply by X-transpose-y. Finally compare the ridge

estimate to the least-squares estimate to interpret shrinkage. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The prior variance is $\tau^2 = 4$, so $\lambda = \sigma^2/\tau^2 = 1$. Thus

$$\hat{\beta} = (X^\top X + I)^{-1} X^\top y = \begin{pmatrix} 5 & 0 \\ 0 & 10 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ 18 \end{pmatrix} = \boxed{\begin{pmatrix} 1.6 \\ 1.8 \end{pmatrix}}.$$

The penalty adds one to each diagonal element of $X^\top X$. The prior is centred at zero, so the estimates are shrunk toward zero relative to the least-squares estimates $(2, 2)^\top$.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.3 Question 3: New practice – LASSO soft thresholding

Original question

In a one-dimensional orthonormal-design problem, the least-squares estimate is 2.4 and the LASSO threshold is 1.5. Compute the LASSO posterior-mode estimate. Repeat if the least-squares estimate is 1.1. Explain the role of the cusp at zero.

Teacher's spoken walkthrough

In the one-dimensional orthonormal case, the LASSO MAP estimate follows the soft-thresholding rule. Subtract the threshold from the magnitude when the magnitude is above the threshold; otherwise set the estimate to zero. Explain that the cusp in the absolute-value penalty allows the optimum to sit exactly at zero. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Soft thresholding is $\operatorname{sign}(z)\max(|z| - \lambda_*, 0)$. For $z = 2.4$ and threshold 1.5, the estimate is $2.4 - 1.5 = \boxed{0.9}$. For $z = 1.1$, the magnitude is below the threshold, so the estimate is $\boxed{0}$. The absolute-value penalty has a sharp corner at zero; the optimum can occur at that corner over a nontrivial range of data values, producing exact zeros in the MAP estimate.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.4 Question 4: New practice – spike-and-slab inclusion

Original question

A predictor has prior inclusion probability $q = 0.20$. The data produce Bayes factor $BF_{10} = 5$ for inclusion versus exclusion. Compute the posterior inclusion probability and explain why this is a direct probability statement about model inclusion.

Teacher's spoken walkthrough

Prior inclusion odds are q divided by one-minus-q. Multiply by the Bayes factor for inclusion relative to exclusion to obtain posterior inclusion odds. Convert those odds to a posterior inclusion probability. Because the model indicator has a discrete prior, the posterior can assign positive probability to inclusion or exclusion. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Prior inclusion odds are $q/(1-q) = 0.2/0.8 = 1/4$. The data multiply these odds by the Bayes factor 5, giving posterior odds $5/4$. Thus

$$\Pr(\gamma = 1 \mid y) = \frac{5/4}{1 + 5/4} = \boxed{5/9 = 0.5556}.$$

The spike-and-slab model has a discrete indicator with positive prior probability on inclusion and exclusion. The posterior inclusion probability is therefore an actual posterior event probability, not merely a statement that a continuous coefficient is close to zero.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.5 Question 5: New practice – posterior predictive regression

Original question

Under a Normal–Inverse-Gamma regression posterior, b_n, V_n, a_n, and d_n are known. Write the posterior predictive distribution for a new response at covariate vector x_*. Identify the location, degrees of freedom, and scale. Explain the two sources of predictive variance.

Teacher's spoken walkthrough

The predictive distribution averages the new-response normal likelihood over the posterior of beta and sigma-squared. Under the Normal–Inverse-Gamma model, the result is a Student-t distribution. State the predictive location, degrees of freedom and scale carefully. Explain that uncertainty in beta and sigma-squared contributes in addition to the new observation noise. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

The posterior predictive distribution is

$$y_* \mid y \sim t_{2a_n}\left(x_*^\top b_n, \sqrt{\frac{d_n}{a_n}(1 + x_*^\top V_n x_*)}\right),$$

where the second argument is the Student-t scale. The location is the posterior mean fitted response $x_*^\top b_n$, degrees of freedom are $2a_n$, and scale squared is $(d_n/a_n)(1 + x_*^\top V_n x_*)$. The 1 represents the variance of a new observation around its conditional mean; the quadratic term represents uncertainty about β. Uncertainty about σ^2 produces the Student-t tails.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.6 Question 6: Book-aligned – ordinary Bayesian linear regression

Original question

With a flat prior on β and known σ^2, derive the posterior mean and covariance of β for a full-rank design matrix X. State the condition required for the posterior covariance to exist.

Teacher's spoken walkthrough

With a flat prior on beta and known variance, the posterior is proportional to the Gaussian likelihood. Complete the square in beta to show that the posterior mean is the least-squares estimate and the covariance is sigma-squared times the inverse of X-transpose-X. The design matrix must have full column rank for this inverse to exist. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

With a flat prior on β and known σ^2, the posterior kernel is

$$\exp\left\{-\frac{(y - X\beta)^\top (y - X\beta)}{2\sigma^2}\right\}.$$

Completing the square yields

$$\boxed{\beta \mid y \sim N\left((X^\top X)^{-1} X^\top y, \sigma^2 (X^\top X)^{-1}\right).}$$

The matrix $X^\top X$ must be invertible, which requires X to have full column rank. If it is rank deficient, a flat prior does not regularise the unidentified directions and this covariance formula does not exist.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.7 Question 7: Book-aligned – prior information as additional data

Original question

Explain how a proper normal prior on the regression coefficients can be represented algebraically as additional quadratic information in the likelihood. Show how the prior precision changes the posterior precision matrix.

Teacher's spoken walkthrough

A normal prior contributes a quadratic penalty to the residual sum of squares. Its precision adds to X-transpose-X, while the prior mean contributes a corresponding linear term. Write the posterior precision and mean. If the prior mean is zero and covariance is tau-squared times identity, the expression reduces to ridge regularisation. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

A normal prior $\beta \sim N(b_0, \sigma^2 V_0)$ adds the quadratic penalty $(\beta - b_0)^\top V_0^{-1}(\beta - b_0)$ to the residual sum of squares on the σ^2 scale. The posterior precision is $V_0^{-1} + X^\top X$ and its mean is $(V_0^{-1} + X^\top X)^{-1}(V_0^{-1} b_0 + X^\top y)$. Thus the prior acts like additional information centred at b_0. If $b_0 = 0$ and $V_0 = \tau^2 I$, the added precision is $\tau^{-2} I$, producing ridge regularisation.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.8 Question 8: Book-aligned – scalar LASSO derivation

Original question

For a one-dimensional coefficient with known variance, write the posterior kernel under a Laplace prior. Derive the soft-thresholding MAP rule from the penalised absolute-error objective. Explain why the multidimensional correlated-predictor case is not a coordinatewise copy of this formula.

Teacher's spoken walkthrough

Write the log posterior as the negative quadratic likelihood term minus lambda times the absolute value of beta. Differentiate separately on the positive and negative sides, then check whether zero itself minimises the objective. The result is the soft-thresholding rule. With correlated predictors, the quadratic likelihood is not separable coefficient by coefficient. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why

the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

In one dimension, with least-squares estimate $\hat{\beta}$ and variance v (known), the likelihood kernel is $\exp\{-(\beta-\hat{\beta})^2/(2v)\}$ and the Laplace prior is $\exp(-\lambda|\beta|)$. The posterior mode minimises

$$\frac{(\beta-\hat{\beta})^2}{2v}+\lambda|\beta|.$$

Its solution is $\operatorname{sign}(\hat{\beta})\max(|\hat{\beta}|-v\lambda,0)$. In the multidimensional case the quadratic term has matrix $X^\top X$; when predictors are correlated, cross-products couple the coordinates, so independent scalar soft thresholds generally do not minimise the joint objective.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

9.6.9 Question 9: Book-aligned – variable selection and model averaging

Original question

For p candidate predictors, define the model indicator vector γ. Write the posterior model probability in terms of prior model probability and marginal likelihood. Explain the difference between selecting the single most probable model and averaging predictions over model uncertainty.

Teacher's spoken walkthrough

The vector gamma identifies which predictors are included. The posterior model probability is proportional to prior model probability times marginal likelihood. A single-model choice selects the model with the greatest posterior probability, while model averaging combines predictions across models weighted by their posterior probabilities. Averaging retains uncertainty about selection. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

For model indicator vector γ, the posterior model probability is

$$\Pr(\gamma \mid y)=\frac{p(y \mid \gamma)\Pr(\gamma)}{\sum_{\gamma'} p(y \mid \gamma')\Pr(\gamma')},$$

where $p(y \mid \gamma)=\int p(y \mid \beta_\gamma,\gamma)p(\beta_\gamma \mid \gamma)\,\mathrm{d}\beta_\gamma$ (and any variance parameters are also integrated out). Selecting the single most probable model reports only $\arg\max_\gamma \Pr(\gamma \mid y)$. Model averaging instead computes predictions such as $p(\tilde{y} \mid y)=\sum_\gamma p(\tilde{y} \mid y,\gamma)\Pr(\gamma \mid y)$, preserving uncertainty about which model is appropriate.

Closing words for the instructor

Close by translating the mathematics back into a modelling statement: which coefficients are shrunk, which can be zero, and what uncertainty is retained? A good answer explains both the algebra and the modelling interpretation.

Connecting question type 6 to type 7

The regression questions considered coefficients within one model. The final type widens the lens: several groups have related parameters, and the model must represent both within-group evidence and between-group variation. This is where hierarchical Bayes, empirical Bayes, and profile likelihood complete the journey.

9.7 Type 7 – hierarchical Bayes, empirical Bayes, and profile likelihood

Opening script for this question type

These questions concern information sharing across groups. Start by drawing the hierarchy, derive the group-level update conditional on hyperparameters, and then explain how the hyperparameters are estimated. We will distinguish full Bayes integration, empirical Bayes plug-in, and profile likelihood maximisation.

9.7.1 Question 1: Fully solved – normal-normal shrinkage

Original question

A group estimate is $\bar{y} = 14$ with sampling variance $s^2 = 1$. Its prior is $\theta \sim N(10, 4)$. Find the posterior mean and variance.

Teacher's spoken walkthrough

This is the basic Normal–Normal update. Think in terms of precision rather than variance. Add the sampling precision to the prior precision to obtain posterior precision. The posterior mean is the precision-weighted average of the observed group estimate and the prior population mean. Then interpret the result as shrinkage toward the population mean. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

The prior precision is $1/4$ and sampling precision is 1. Posterior precision is $1 + 1/4 = 5/4$, so posterior variance is $4/5 = 0.8$. The posterior mean is

$$m = 0.8\,(14) + 0.2\,(10) = \boxed{13.2}.$$

Thus $\theta \mid y \sim N(13.2, 0.8)$. The group estimate is pulled from 14 toward the population mean 10. The group data receive weight $\tau^2/(\tau^2 + s^2) = 4/5 = 0.8$.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.2 Question 2: Fully solved – comparing shrinkage across groups

Original question

Three group estimates are $\bar{y} = (8, 10, 16)$ with sampling variances $s_j^2 = (1, 4, 1)$. Conditional on known hyperparameters, $\theta_j \sim N(11, 3)$. Find the posterior mean for each group.

Teacher's spoken walkthrough

Use the same shrinkage formula for all three groups. Calculate a separate weight for each group using its sampling variance. Groups one and three have equal precision and equal weights; group two is less precise, so its observed mean receives less weight. Explain that stronger shrinkage does not mean the group estimate is ignored; it means the population distribution contributes more information. Now write the solution line by line. Do not jump to the final number. At each line, say which probability rule or algebraic identity is being used, and explain why the terms can be combined in that way. After the calculation, return to the wording of the question and state the answer in the requested form.

Worked reasoning and final answer

For each group, $w_j = 3/(3 + s_j^2)$ and $m_j = w_j \bar{y}_j + (1 - w_j)11$. For groups 1 and 3, $w = 3/4$:

$$m_1 = \tfrac{3}{4}(8) + \tfrac{1}{4}(11) = 8.75, \qquad m_3 = \tfrac{3}{4}(16) + \tfrac{1}{4}(11) = 14.75.$$

For group 2, $w = 3/7$:

$$m_2 = \tfrac{3}{7}(10) + \tfrac{4}{7}(11) = \frac{74}{7} = 10.5714.$$

The second group has the largest sampling variance and is shrunk most strongly toward 11. The posterior variance is $v_j = (1/s_j^2 + 1/3)^{-1}$ for each group.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.3 Question 3: New practice – posterior for a group parameter

Original question

In the normal-normal model, derive the conditional posterior variance and mean of θ_j given $\bar{y}_j$, μ, and τ^2. State how the posterior changes as s_j^2 increases and as τ^2 decreases.

Teacher's spoken walkthrough

Derive the conditional posterior variance as the reciprocal of the sum of precisions. The posterior mean uses the same precision weights. As sampling variance increases, the group data become less precise and shrinkage toward mu becomes stronger. As tau-squared decreases, the population distribution concentrates more tightly and also produces stronger pooling. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Multiplication of the normal sampling density and normal prior gives posterior precision $1/s_j^2+1/\tau^2$ and precision-weighted mean. Therefore

$$v_j = \left(\frac{1}{s_j^2} + \frac{1}{\tau^2}\right)^{-1}, \qquad m_j = v_j\left(\frac{\bar{y}_j}{s_j^2} + \frac{\mu}{\tau^2}\right).$$

Equivalently, $m_j = w_j\bar{y}_j + (1-w_j)\mu$ with $w_j = \tau^2/(\tau^2+s_j^2)$. Increasing s_j^2 reduces the weight on the noisy group estimate and increases shrinkage. Decreasing τ^2 says that group effects are more similar in the population, so the prior precision $1/\tau^2$ increases and shrinkage toward μ becomes stronger.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.4 Question 4: New practice – empirical Bayes by moments

Original question

Suppose $Y_j \sim N(\mu, s^2+\tau^2)$ independently with known common sampling variance s^2. Derive method-of-moments estimators for μ and τ^2 from the J group estimates. Explain why a nonnegative constraint on τ^2 is necessary.

Teacher's spoken walkthrough

The marginal distribution of each observed group estimate has mean mu and variance s-squared plus tau-squared. Use the sample mean to estimate mu, and use the observed variance across groups to estimate tau-squared after subtracting known within-group variance. The estimate must be constrained to be nonnegative because a variance cannot be negative. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Since $Y_j \sim N(\mu, s^2+\tau^2)$,

$$\mathbb{E}(Y_j) = \mu, \qquad \mathrm{Var}(Y_j) = s^2+\tau^2.$$

The method-of-moments estimator of μ is the average of group estimates, $\hat{\mu} = \bar{Y}$. Equating the empirical second central moment to the model variance gives

$$\hat{\tau}^2_{MOM} = \frac{1}{J}\sum_{j=1}^{J}(Y_j - \bar{Y})^2 - s^2.$$

Because a variance component cannot be negative, use $\hat{\tau}^2 = \max(0, \hat{\tau}^2_{MOM})$. If empirical variation between group estimates is smaller than the known sampling variance, the data do not support a positive additional between-group variance under this simple model.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.5 Question 5: New practice – profile likelihood

Original question

For the same model $Y_j \sim N(\mu, s^2 + \tau^2)$, derive the profile log-likelihood for τ^2 by maximising over μ. Obtain the constrained maximum-likelihood estimate of τ^2 and explain why the divisor is J rather than $J - 1$.

Teacher's spoken walkthrough

For each fixed tau-squared, maximise the normal likelihood over mu. The maximiser is the average of the group estimates. Substitute it into the log likelihood to obtain the profile function. The unconstrained variance-component estimate is the observed maximum-likelihood variance minus s-squared; if this is negative, the constrained estimate is zero. The divisor is J because we are deriving a maximum-likelihood estimate, not the unbiased sample variance. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Let $v = s^2 + \tau^2$. The log-likelihood is

$$\ell(\mu, \tau^2) = -\frac{J}{2}\log v - \frac{1}{2v}\sum_j (Y_j - \mu)^2 + \text{constant}.$$

For each fixed τ^2, differentiation with respect to μ gives $\hat{\mu}(\tau^2) = \bar{Y}$. Substitution yields

$$\ell_p(\tau^2) = -\frac{J}{2}\log(s^2 + \tau^2) - \frac{SS}{2(s^2 + \tau^2)} + C, \quad SS = \sum_j (Y_j - \bar{Y})^2.$$

Differentiate with respect to v: $\partial \ell_p / \partial v = -J/(2v) + SS/(2v^2)$. The unconstrained maximum is $\hat{v} = SS/J$, hence $\hat{\tau}^2 = SS/J - s^2$. Applying $\tau^2 \geq 0$ gives

$$\boxed{\hat{\tau}^2 = \max\{0, SS/J - s^2\}.}$$

The denominator is J because the likelihood maximiser of a normal variance uses the maximum-likelihood divisor. Dividing by $J-1$ instead gives the unbiased sample variance, which solves a different estimation objective.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.6 Question 6: Book-aligned – Beta–Binomial hierarchy

Original question

Suppose $Y_j \mid p_j \sim \text{Binomial}(n_j, p_j)$ and $p_j \mid a, b \sim \text{Beta}(a, b)$. Derive the group-level posterior $p_j \mid y_j, a, b$ and the marginal likelihood contribution $p(y_j \mid a, b)$. Explain how the shared hyperparameters link the groups.

Teacher's spoken walkthrough

Given a and b, update each group's Beta parameters by adding its successes and failures. Then integrate p-j out to obtain a Beta–Binomial marginal contribution. Multiplying these contributions over groups gives the marginal likelihood for the shared hyperparameters. These shared parameters connect the groups and allow information to pass between them. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Given a, b, conjugacy gives

$$p_j \mid y_j, a, b \sim \text{Beta}(a + y_j, b + n_j - y_j).$$

Integrating p_j from the Binomial likelihood times Beta prior gives

$$p(y_j \mid a, b) = \binom{n_j}{y_j} \frac{B(a + y_j, b + n_j - y_j)}{B(a, b)}.$$

The joint marginal likelihood is the product over groups. The common a, b determine the population distribution of the success probabilities, so observations in one group affect posterior learning about a, b and indirectly influence estimates for other groups. This is the source of partial pooling.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.7 Question 7: Book-aligned – exchangeability and pooling

Original question

Define exchangeability for group parameters. Explain no pooling, complete pooling, and partial pooling, and discuss which group estimates are shrunk most strongly in a normal hierarchical model.

Teacher's spoken walkthrough

Define exchangeability as invariance under relabelling groups. Then contrast the three pooling strategies. No pooling keeps each group isolated. Complete pooling forces a common value. Partial pooling allows group variation but connects estimates through a shared population distribution. The least precise groups are generally shrunk the most. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Exchangeability means that the joint prior distribution of $(\theta_1, \ldots, \theta_J)$ is unchanged under permutations of group labels. No pooling estimates each θ_j using only its own group's data. Complete pooling sets all group parameters equal. Partial pooling permits variation but draws group parameters from a shared population distribution. Under the normal-normal model, groups with larger sampling variances (less precise estimates) receive smaller weights on their own sample means and are shrunk more strongly toward the population mean. Groups with precise data are shrunk less.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.8 Question 8: Book-aligned – parallel experiments and predictive inference

Original question

Several experiments estimate related normal means. Explain how the eight-schools-style hierarchy produces estimates between the unpooled observations and the overall mean. State what happens as the between-group variance tends to zero or infinity.

Teacher's spoken walkthrough

The eight-schools example has a noisy estimate from each school and a common population distribution for the true effects. Each posterior group estimate is a weighted average of its own observed estimate and the population mean. As the between-group variance tends to zero, pooling approaches complete pooling. As it grows large, estimates approach the unpooled values. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

In an eight-schools-style model, each school's observed estimate is noisy with known sampling standard error, while the true school effect is drawn from a common normal population with mean μ and variance τ^2. Conditional on these hyperparameters, the posterior mean for each school is a precision-weighted average of its observed estimate and μ. As $\tau^2 \to 0$, all school effects are concentrated near the same population mean and pooling becomes nearly complete. As $\tau^2 \to \infty$, the prior becomes diffuse relative to sampling errors and each school estimate is close to its unpooled value.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

9.7.9 Question 9: Book-aligned – full Bayes versus empirical Bayes

Original question

Compare full hierarchical Bayes and empirical Bayes. Write the posterior for a group parameter under each approach and discuss the consequences of plugging in a point estimate of the hyperparameters. Explain how profile likelihood differs from integrating out nuisance parameters.

Teacher's spoken walkthrough

Full Bayes integrates over the posterior distribution of the hyperparameters; empirical Bayes plugs in an estimated hyperparameter value. This difference affects uncertainty. Then compare profiling with Bayesian integration: profiling maximises over nuisance parameters, while marginalisation averages over them using a probability distribution. They are different operations, even when they concern the same nuisance parameters. As you present the answer, keep the question's requested quantity in view. A derivation earns marks because it shows why the stated posterior or estimator follows from the model. Read each line, identify the role of the terms, and finish with a sentence interpreting the result rather than ending at an unexplained formula.

Worked reasoning and final answer

Full hierarchical Bayes assigns a hyperprior $p(\eta)$ and obtains

$$p(\theta_j \mid y) = \int p(\theta_j \mid \eta, y) p(\eta \mid y) \, \mathrm{d}\eta.$$

Empirical Bayes estimates $\hat{\eta}$ from the marginal likelihood and plugs it in, giving $p_{EB}(\theta_j \mid y) = p(\theta_j \mid \hat{\eta}, y)$. Plugging in ignores uncertainty about η and can yield overconfident group-level intervals, especially with few groups. Profile likelihood maximises over nuisance parameters for each fixed parameter of interest; Bayesian marginalisation integrates over nuisance parameters using a probability distribution. Maximisation and integration are different operations and need not produce the same estimate or uncertainty.

Closing words for the instructor

End by explaining where the information is coming from: the group's own data, the shared population distribution, or the hyperparameter estimation method. Ask the class which uncertainty is retained and which is ignored under each approach.

The final story: from one positive test to a population-level scientific conclusion

A continuous closing lecture

Let us finish by returning to the public-health team with which we began. At the start, the team saw a positive diagnostic result and wanted to know what it meant. We learned that the answer depends on both the accuracy of the test and the prevalence of disease. That was our first lesson: evidence must be interpreted conditionally, and the direction of conditioning matters.

The team then collected many observations. Some were binary outcomes, some were counts, some were waiting times, and some were categorical outcomes. Exponential-family models gave us a common language for these data. Conjugate priors allowed us to update systematically, often by adding sample information to prior hyperparameters. We learned that the algebra is not magic: the posterior appears because the prior kernel and likelihood kernel combine into a recognisable probability density.

Once the posterior had been obtained, the team needed to communicate it. We learned that a point estimate is a summary, not the whole answer. The posterior mean, median, and mode can differ because they answer different questions. Credible intervals describe posterior uncertainty. Prior choice matters, and Jeffreys' prior gives a principled construction based on the information in the likelihood, while still requiring care about propriety and parameterisation.

The scientific team next had to compare two explanations: perhaps an intervention changes disease risk, or perhaps it does not. We learned that posterior odds update prior odds through a Bayes factor. The Bayes factor compares marginal likelihoods, so it rewards models that assign predictive probability to the observed data rather than simply rewarding a model for fitting well after seeing them.

As the model grew more complicated, direct integration became impractical. MCMC allowed us to explore the posterior through dependent simulation. Gibbs sampling used full conditional distributions; Metropolis–Hastings used proposal moves and an acceptance probability. We also learned a crucial lesson: computation does not certify itself. Trace plots, autocorrelation plots, multiple chains, effective sample size, and other diagnostics are part of the statistical argument.

The team then asked why disease risk differed across individuals. Bayesian linear regression connected the outcome to several predictors. The Normal–Inverse-Gamma prior gave a conjugate structure. A Gaussian prior led to ridge-type shrinkage; a Laplace prior led to the LASSO posterior mode; and spike-and-slab priors represented uncertainty about whether a predictor should be included at all. These are not unrelated tricks. They are different prior structures expressing different beliefs about coefficient size and variable inclusion.

Finally, the team recognised that individuals belonged to districts. It would be unreasonable to ignore all similarities between districts, but it would also be unreasonable to force every district to have the same parameter. Hierarchical models created partial pooling. Full Bayes retained uncertainty about population-level hyperparameters, while empirical Bayes estimated them from the data. Profile likelihood offered a different operation: maximisation over nuisance parameters rather than Bayesian integration.

That is the entire journey. We began with one uncertain event and ended with a structured

model for many related populations. The same reasoning has carried us throughout: define the unknown quantity, specify how data arise, represent prior information, condition on observed evidence, derive or simulate the posterior, summarise uncertainty, compare explanations when needed, and interpret the result in the language of the scientific problem.
If you remember only one sentence from this course, remember this: Bayesian inference is not a collection of formulas for updating distributions; it is a coherent way of reasoning from assumptions and evidence to uncertainty-aware conclusions. The formulas are the tools. The model is the argument. The interpretation is the scientific answer.

Closing lecture: how to approach an unfamiliar Bayesian examination problem

Final words for the instructor

When you receive a Bayesian question in the examination, do not begin by searching your memory for a final formula. Begin by identifying the unknown quantity and the sampling model. Write the likelihood conditional on the unknown parameter. Then write the prior, making sure its support and parameterisation match the problem. Multiply the two kernels and simplify. If the posterior is conjugate, identify the family and its updated parameters. If it is not conjugate, describe the computational method that would produce posterior summaries.
Next, answer exactly what the examiner asked for. If the question asks for a posterior mean, calculate the mean. If it asks for a MAP estimate, maximise the posterior density. If it asks for a credible interval, identify the posterior quantiles or HPD region. If it asks for a Bayes factor, integrate the likelihood under each model before taking the ratio. If it asks for an MCMC diagnostic, interpret the trace, density, and autocorrelation plots rather than simply listing them.
Finally, interpret the answer. A probability should be stated in words, a shrinkage estimate should be compared with the unpooled estimate, and a model comparison should identify which model is favoured. The best examination solutions are not merely correct calculations. They are complete statistical arguments in which every formula has a clear role.

Primary sources and relationship to the companion guide

This speech manual follows the chapter sequence, section headings, formula conventions, question statements, and answer key of *Bayesian_Inference_Endsem_Guide.tex*. The primary textbook alignment stated in that guide is Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin, *Bayesian Data Analysis*, Third Edition. The guide identifies the principal textbook coverage as Chapter 1 for Bayesian foundations; Chapters 2–3 for conjugate models and posterior summaries; Chapter 5 for hierarchical models; Section 7.4 for Bayes factors; Chapter 11 for Gibbs, Metropolis–Hastings, and convergence assessment; Chapter 14 for Bayesian regression and regularisation; and Chapter 20 for mixture priors and variable selection.

The profile-likelihood explanation is retained as a syllabus connection, with the distinction from Bayesian marginalisation made explicit. This manual is an original spoken teaching companion; it does not reproduce the textbook's prose.